

1 Nonparametric estimation

Exercise 1.1. Discuss the behavior of a Nadaraya-Watson mean regression estimate when one of i.i.d. observations, say (x_1, y_1) , tends to assume extreme values. Specifically, discuss

- (a) the case when x_1 is very big,
- (b) the case when y_1 is very big.

Solution.

- (a) When x_1 is very big, we will have a big influence of this observation when estimating the regression in the right region of the support. With bounded kernels, we may find that no observations fall into the window. With unbounded kernels, the estimated regression line will go almost through (x_1, y_1) .
- (b) When y_1 is very big, the estimated regression line will have a hump in the region centered at x_1 .

Exercise 1.2. Firms produce some product using technology $f(l, k)$. The functional form of f is unknown, although we know that it exhibits constant returns to scale. For a firm i , we observe labor l_i , capital k_i , and output y_i , and the data-generating process takes the form $y_i = f(l_i, k_i) + \varepsilon_i$, where $\mathbb{E}[\varepsilon_i] = 0$ and ε_i is independent of (l_i, k_i) . Using random sample $\{y_i, k_i, l_i\}_{i=1}^n$ suggest a nonparametric estimator of $f(l, k)$ which also exhibits constant returns to scale.

Solution. The CRS technology has the property that

$$f(l, k) = kf\left(\frac{l}{k}, 1\right).$$

Hence, the data-generating process is equivalent to

$$\frac{y_i}{k_i} = f\left(\frac{l_i}{k_i}, 1\right) + \frac{\varepsilon_i}{k_i},$$

from which follows that the Nadaraya-Watson estimator can be written as

$$\hat{f}\left(\frac{l}{k}, 1\right) = \frac{\sum_{i=1}^n K_h\left(\frac{l_i}{k_i} - \frac{l}{k}\right) \frac{y_i}{k_i}}{\sum_{i=1}^n K_h\left(\frac{l_i}{k_i} - \frac{l}{k}\right)} \Rightarrow \hat{f}(l, k) = k \cdot \frac{\sum_{i=1}^n K_h\left(\frac{l_i}{k_i} - \frac{l}{k}\right) \frac{y_i}{k_i}}{\sum_{i=1}^n K_h\left(\frac{l_i}{k_i} - \frac{l}{k}\right)}.$$

Exercise 1.3. Recall the Nadaraya-Watson kernel estimator $\hat{g}(x)$ of the conditional mean $g(x) := \mathbb{E}[y|x]$ constructed for a random sample. Show that if $g(x) = c$, where c is some constant, then $\hat{g}(x)$ is unbiased.

Solution. Assume having a random sample $x_1, \dots, x_n$, and collect it as $\mathcal{X}$. Recall that

$$\hat{g}(x) = \frac{\sum_{i=1}^n y_i K_h(x_i - x)}{\sum_{i=1}^n K_h(x_i - x)}.$$

Then,

$$\begin{aligned} \mathbb{E}[\hat{g}(x)] &= \mathbb{E}[\mathbb{E}[\hat{g}(x)|\mathcal{X}]] \\ &= \mathbb{E}\left[\mathbb{E}\left[\frac{\sum_{i=1}^n y_i K_h(x_i - x)}{\sum_{i=1}^n K_h(x_i - x)} \middle| \mathcal{X}\right]\right] \\ &= \mathbb{E}\left[\frac{\sum_{i=1}^n \mathbb{E}[y_i|\mathcal{X}] K_h(x_i - x)}{\sum_{i=1}^n K_h(x_i - x)}\right] \\ &= c \mathbb{E}\left[\frac{\sum_{i=1}^n K_h(x_i - x)}{\sum_{i=1}^n K_h(x_i - x)}\right] = c. \end{aligned}$$

First equality uses the law of iterated expectations, the second uses the definition of the $\hat{g}(x)$, the third follows because kernels are functions of $\mathcal{X}$, the fourth follows from the definition of the conditional mean.

Exercise 1.4. Suppose we have a random sample $\{x_i\}_{i=1}^n$ and let

$$\hat{F}(x) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{x_i \leq x\}$$

denote the empirical distribution function of x_i . Consider two density estimators:

- one-sided estimator:

$$\hat{f}_1(x) = \frac{\hat{F}(x+h) - \hat{F}(x)}{h},$$

- two-sided estimator:

$$\hat{f}_2(x) = \frac{\hat{F}(x+h/2) - \hat{F}(x-h/2)}{h}.$$

Show that:

- $\hat{F}(x)$ is an unbiased estimator of $F(x)$.
- The bias of $\hat{f}_1(x)$ is $O(h^a)$. Find the value of a .
- The bias of $\hat{f}_2(x)$ is $O(h^b)$. Find the value of b .

Solution. Please, consult the note on “big-o”, “small-o” notation in case derivations are unclear.

- We show the unbiasedness using the fact that $\mathbb{E}[\mathbb{1}\{x_i \leq x\}] = F(x)$. Intuition is as follows: recall that $\mathbb{E}[\mathbb{1}\{x_i = x\}] = \mathbb{P}\{x_i = x\}$; intuitively, with inequality instead of equality those probabilities are summing up until the point x , which leads to equivalence with the cumulative distribution function.

Now, the unbiasedness,

$$\mathbb{E}[\hat{F}(x)] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n \mathbb{1}\{x_i \leq x\}\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[\mathbb{1}\{x_i \leq x\}] = F(x).$$

- We will use the following Taylor approximation of $F(x+h)$ around x :

$$F(x+h) = F(x) + hf'(x) + \frac{1}{2}h^2 f''(x) + o(h^2).$$

Substituting it, we have the bias

$$\begin{aligned} \mathbb{E}[\hat{f}_1(x) - f(x)] &= \frac{1}{h} (F(x+h) - F(x)) - f(x) \\ &= \frac{1}{h} \left(F(x) + hf'(x) + \frac{1}{2}h^2 f''(x) + o(h^2) - F(x) \right) - f(x) \\ &= \frac{1}{2}hf''(x) + o(h) \Rightarrow a = 1. \end{aligned}$$

(c) Similarly to the point (b), we expand $F\left(x + \frac{h}{2}\right)$ and $F\left(x - \frac{h}{2}\right)$ around x :

$$F\left(x + \frac{h}{2}\right) = F(x) + \frac{h}{2}f(x) + \frac{1}{2}\left(\frac{h}{2}\right)^2 f'(x) + \frac{1}{6}\left(\frac{h}{2}\right)^3 f''(x) + o(h^3),$$

and

$$F\left(x - \frac{h}{2}\right) = F(x) - \frac{h}{2}f(x) + \frac{1}{2}\left(\frac{h}{2}\right)^2 f'(x) - \frac{1}{6}\left(\frac{h}{2}\right)^3 f''(x) + o(h^3).$$

Then,

$$\begin{aligned} \mathbb{E}[\hat{f}_2(x)] - f(x) &= \frac{1}{h} \left(F\left(x + \frac{h}{2}\right) - F\left(x - \frac{h}{2}\right) \right) - f(x) \\ &= \frac{1}{h} (F(x) + \dots + o(h^3) - F(x) + \dots + o(h^3)) \\ &= \frac{1}{h} \left(\frac{2}{6} \left(\frac{h}{2}\right)^3 f''(x) \right) + o(h^2) \\ &= \frac{1}{24} h^2 f''(x) + o(h^2) \Rightarrow b = 2. \end{aligned}$$

Exercise 1.5. Derive the asymptotic distribution of the Nadaraya-Watson estimator of the density of a scalar random variable x having a continuous distribution, similarly to how the asymptotic distribution of the Nadaraya-Watson estimator of the regression function is derived, under similar conditions. Give interpretation to how your expressions for asymptotic bias and asymptotic variance depend on the shape of the density.

Solution. Recall the Nadaraya-Watson density estimator, $\hat{f}(x) = \frac{1}{n} \sum_{i=1}^n K_h(x_i - x)$. During the first lecture you showed that $\mathbb{E}[\hat{f}(x)] = f(x) + o(1)$ as $h \rightarrow 0$. However, this result is not particularly revealing, since we do not see what causes the bias as h approaches (but not yet equal to) zero. To have a better picture, we work on the bias further taking a higher-order expansion of the density around x (the point we are estimating the density function at).

Now,

$$\begin{aligned} \mathbb{E}[\hat{f}(x) - f(x)] &= \int \frac{1}{h} K\left(\frac{x_i - x}{h}\right) f(x_i) dx_i - f(x) \\ &\stackrel{\{u = \frac{x_i - x}{h}\}}{=} \int K(u) f(x + hu) du - f(x) \\ &= \int K(u) \left(f(x) + f'(x)hu + \frac{1}{2}f''(x)h^2u^2 + o(h^2) \right) du - f(x) \\ &= f(x) \int K(u) du + f'(x)h \int uK(u) du + \frac{1}{2}f''(x)h^2 \int u^2 K(u) du + o(h^2) - f(x) \\ &= \frac{1}{2}f''(x)h^2 \sigma_K^2 + o(h^2). \end{aligned}$$

In particular, we use here the fact that $\int K(u) du = 1$ due to normalization of the kernel, and $\int uK(u) du = 0$ due to the kernel symmetry.

What we see here is revealing: the bias is proportional to the second derivative of the density function. The second derivative, which describes the curvature of the density, is negative ($f''(x) < 0$) around the mode of the distribution, and is positive ($f''(x) > 0$) in the tails. That means that the density estimator will be downward-biased around the mode, and upward-biased around the tails. Also note that our results do not contradict the results you have obtained during the lecture: indeed, by putting $h \rightarrow 0$, we are getting rid of the bias.

We start computing the variance with

$$\begin{aligned} \text{var}[\hat{f}(x)] &= \text{var}\left[\frac{1}{n} \sum_{i=1}^n K_h(x_i - x)\right] = \text{var}\left[\frac{1}{n} \sum_{i=1}^n \frac{1}{h} K\left(\frac{x_i - x}{h}\right)\right] \\ &= \frac{1}{n^2 h^2} \text{var}\left[\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right)\right] = \frac{n}{n^2 h^2} \text{var}\left[K\left(\frac{x_i - x}{h}\right)\right] = \frac{1}{n h^2} \text{var}\left[K\left(\frac{x_i - x}{h}\right)\right] \end{aligned}$$

where the forth equality is implied by the i.i.d. assumption. Next I will normalize by nh to get

$$\begin{aligned}
nh \text{var} \left[\hat{f}(x) \right] &= \frac{1}{h} \text{var} \left[K \left(\frac{x_i - x}{h} \right) \right] \\
&= \frac{1}{h} \mathbb{E} \left[K \left(\frac{x_i - x}{h} \right)^2 \right] - \frac{1}{h} \left(\mathbb{E} \left[K \left(\frac{x_i - x}{h} \right) \right] \right)^2 \\
&= \frac{1}{h} \mathbb{E} \left[K \left(\frac{x_i - x}{h} \right)^2 \right] - \frac{1}{h} \left(h \cdot \frac{1}{h} \mathbb{E} \left[K \left(\frac{x_i - x}{h} \right) \right] \right)^2 \\
&= \frac{1}{h} \mathbb{E} \left[K \left(\frac{x_i - x}{h} \right)^2 \right] - h \left(\mathbb{E} \left[\frac{1}{h} K \left(\frac{x_i - x}{h} \right) \right] \right)^2.
\end{aligned}$$

Note that we have already derived the second term (that is squared here). Specifically,

$$\mathbb{E} \left[\hat{f}(x) \right] = \mathbb{E} \left[\frac{1}{h} K \left(\frac{x_i - x}{h} \right) \right] = f(x) + \frac{1}{2} h^2 f''(x) + o(h^2). \quad (1)$$

Now, let's work with the first term. We have

$$\begin{aligned}
\frac{1}{h} \mathbb{E} \left[K \left(\frac{x_i - x}{h} \right)^2 \right] &= \frac{1}{h} \int K \left(\frac{x_i - x}{h} \right)^2 f(x_i) dx_i \\
&\stackrel{\{u = \frac{x_i - x}{h}\}}{=} \int K(u)^2 f(x + hu) du \\
&= \int K(u)^2 (f(x) + o(1)) du \\
&= f(x) \int K(u)^2 du + o(1) \\
&= f(x) R_K + o(1).
\end{aligned}$$

Putting everything together, we have

$$\begin{aligned}
nh \text{var} \left[\hat{f}(x) \right] &= \frac{1}{h} \mathbb{E} \left[K \left(\frac{x_i - x}{h} \right)^2 \right] - h \left(\mathbb{E} \left[\frac{1}{h} K \left(\frac{x_i - x}{h} \right) \right] \right)^2 \\
&= f(x) R_K + o(1) - h \left(f(x) + \frac{1}{2} h^2 f''(x) + o(h^2) \right)^2 \\
&= f(x) R_K + o(1) - \left(h^2 f(x) + \frac{1}{2} h^4 f''(x) + o(h^4) \right)^2 \\
&= f(x) R_K + o(1) - (\mathcal{O}(h^2))^2 \\
&= f(x) R_K + o(1) - \mathcal{O}(h^4) \\
&= f(x) R_K + o(1)
\end{aligned}$$

where we denote $\mathcal{O}(h^2) = h^2 f(x) + \frac{1}{2} h^4 f''(x) + o(h^4)$, because indeed by the definition of "big-o",

$$\mathcal{O}(h^2) = \frac{h^2 f(x) + \frac{1}{2} h^4 f''(x) + o(h^4)}{h^2} = f(x) + \frac{1}{2} h^2 f''(x) + o(h^2) \xrightarrow{h \rightarrow 0} f(x) < \infty.$$

Dividing $nh \text{var} \left[\hat{f}(x) \right]$ by nh , we obtain the variance of the Nadaraya-Watson density estimator,

$$\text{var} \left[\hat{f}(x) \right] = \frac{f(x) R_K}{nh} + o \left(\frac{1}{nh} \right) = \mathcal{O} \left(\frac{1}{nh} \right).$$

The CLT of the estimator is implied by the Linderberg CLT for heterogeneous random variables (for which we should verify the Linderberg condition; for simplicity we do not, but it does hold). Hence,

$$\sqrt{nh} \left(\hat{f}(x) - f(x) \right) \xrightarrow{d} \mathcal{N} \left(\frac{1}{2} \lambda f''(x), f(x) R_K \right)$$

where $\lambda := \lim_{n \rightarrow \infty} \sqrt{nh^5} < \infty \Leftrightarrow h = \mathcal{O}(n^{-1/5})$. Notice that the convergence rate should correspond *inverse-proportionally to the variance*. Remember, that $h \rightarrow 0$ in the limit so we should “delete” h from the variance to stabilize the asymptotic distribution. For example, if the convergence rate is $\sqrt{nh^2}$, we have $hf(x)R_K \rightarrow 0$ as $h \rightarrow 0$. If, on the other hand, the convergence rate is $\sqrt{nh^{-1}}$, $\frac{f(x)R_K}{h^3} \rightarrow \infty$ as $h \rightarrow 0$. The convergence rate $\sqrt{nh}$ perfectly balances those two extreme cases giving us a possibility of analyzing the non-degenerate distribution.

Exercise 1.6. Analyze carefully the asymptotic properties of the Nadaraya-Watson estimator of a regression function with perfect fit, i.e., when the variance of the error is zero.

Solution. Given the setup, we can infer that uncertainty will be coming from the distribution of x_i only, since $y_i = g(x_i)$. Let's write the Nadaraya-Watson regression estimator, plug in the data-generating process for y_i , and rewrite:

$$\begin{aligned}\hat{g}(x) &= \frac{\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right) y_i}{\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right)} = \frac{\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right) g(x_i)}{\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right)} \\ &\Rightarrow \hat{g}(x) - g(x) = \frac{\frac{1}{nh} \sum_{i=1}^n K\left(\frac{x_i - x}{h}\right) (g(x_i) - g(x))}{\frac{1}{nh} \sum_{i=1}^n K\left(\frac{x_i - x}{h}\right)}.\end{aligned}$$

We already know that the denominator is the Nadaraya-Watson kernel density estimator. Hence, by previous results,

$$\hat{f}(x) := \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x_i - x}{h}\right) \xrightarrow{p} f(x) \neq 0.$$

Given that the probability limit of the denominator is not zero, it is sufficient to show that the numerator is not zero either to have a tractable limiting distribution.

We now focus on the nominator, which is, however, equal to $\hat{q}_2(x)$ term that we have worked with before (during lectures). Specifically, with

$$\hat{q}_2(x) := \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x_i - x}{h}\right) (g(x_i) - g(x)) \quad (2)$$

you have showed that

$$\mathbb{E}[\hat{q}_2(x)] = h^2 B(x) f(x) \sigma_K^2 + o(h^2), \quad \text{var}[\hat{q}_2(x)] = o\left(\frac{1}{nh}\right),$$

with $B(x) := \frac{g''(x)}{2} + \frac{g'(x)f'(x)}{f(x)}$. Equation (2) is the average of random variables

$$\zeta_i(x) := \frac{1}{h} K\left(\frac{x_i - x}{h}\right) (g(x_i) - g(x)).$$

We now focus on the expectation and the variance of $\zeta_i(x)$ to derive its distribution, and once we have it, we could derive the distribution of $\hat{g}(x)$ as well.

Now,

$$\begin{aligned}
\mathbb{E} [\zeta_i(x)^2] &= \frac{1}{h^2} \int K \left(\frac{x_i - x}{h} \right)^2 (g(x_i) - g(x))^2 f(x_i) dx_i \\
&\stackrel{\{u = \frac{x_i - x}{h}\}}{=} \frac{1}{h} \int K(u)^2 (g(x + hu) - g(x))^2 f(x + hu) du \\
&= \frac{1}{h} \int K(u)^2 (g'(x)hu + o(h))^2 (f(x) + o(1)) du \\
&= \frac{1}{h} \int (g'(x)hu + o(h))^2 K(u)^2 f(x) du + o(h) \\
&= hg'(x)^2 f(x) \int u^2 K(u)^2 du + o(h) \\
&= hg'(x)^2 f(x) \Psi_K^2 + o(h)
\end{aligned}$$

with $\Psi_K^2 := \int u^2 K(u)^2 du$. Similarly,

$$\begin{aligned}
\mathbb{E} [\zeta_i(x)] &= \int \frac{1}{h} K \left(\frac{x_i - x}{h} \right) (g(x_i) - g(x)) f(x_i) dx_i \\
&\stackrel{\{u = \frac{x_i - x}{h}\}}{=} \int K(u) (g(x + hu) - g(x)) f(x + hu) du \\
&= \int K(u) \cdot o(1)(f(x) + f'(x)hu) du = o(h).
\end{aligned}$$

Together, the variance is $\text{var} [\zeta_i(x)] = \mathbb{E} [\zeta_i(x)^2] - \mathbb{E} [\zeta_i(x)]^2$ or

$$hg'(x)^2 f(x) \Psi_K^2 + o(h) + o(h^2) = hg'(x)^2 f(x) \Psi_K^2 + o(h).$$

By some form of the CLT (notice the change of the convergence rate), we obtain the limiting distribution of ζ_i ,

$$\sqrt{nh^{-1}} \left(\frac{1}{nh} \sum_{i=1}^n K \left(\frac{x_i - x}{h} \right) (g(x_i) - g(x)) - h^2 B(x) \sigma_K^2 f(x) + o(h^2) \right) \xrightarrow{d} \mathcal{N} (0, g'(x)^2 f(x) \Psi_K^2).$$

We are finally ready to derive the limiting distribution of $\hat{g}(x)$. So by the CLT and Slutsky's theorem,

$$\sqrt{nh^{-1}}(\hat{g}(x) - g(x)) \xrightarrow{d} \mathcal{N} \left(\phi \sigma_K^2 B(x), \frac{g'(x)^2}{f(x)} \Psi_K^2 \right)$$

with $\phi := \lim_{n \rightarrow \infty} \sqrt{nh^3}$.

Exercise 1.7. This problem illustrates the use of a difference operator in nonparametric estimation with i.i.d. data. Suppose there is a scalar variable z that takes values on a bounded support. For simplicity, let z be deterministic and compose a uniform grid on the unit interval $[0, 1]$. The other variables are i.i.d. Assume that for the function $g(\cdot)$ below the following Lipschitz condition is satisfied:

$$|g(u) - g(v)| \leq G |u - v|$$

for some constant G .

1. Consider a nonparametric regression of y on z :

$$y_i = g(z_i) + e_i, \quad i = 1, \dots, n,$$

where $\mathbb{E}[e_i | z_i] = 0$. Let the data $\{(z_i, y_i)\}_{i=1}^n$ be ordered so that the z s are in increasing order. A first difference transformation results in the following set of equations:

$$y_i - y_{i-1} = g(z_i) - g(z_{i-1}) + e_i - e_{i-1}, \quad i = 2, \dots, n. \quad (3)$$

The target is to estimate $\sigma^2 := \mathbb{E}[e_i^2]$. Propose its consistent estimator based on the first-difference transformed regression (10). Prove consistency of your estimator.

2. Consider the following partially linear regression of y on x and z :

$$y_i = x'_i \beta + g(z_i) + e_i, \quad i = 1, \dots, n, \quad (4)$$

where $\mathbb{E}[e_i | x_i, z_i] = 0$. Let the data $\{x_i, z_i, y_i\}_{i=1}^n$ be ordered so that z s are in increasing order. The target is to nonparametrically estimate $g(z)$. Propose its consistent estimator using the first-difference transformation of (11).

Solution.

1. Start with the squared average,

$$\begin{aligned} \frac{1}{n-1} \sum_{i=2}^n (y_i - y_{i-1})^2 &= \frac{1}{n-1} \sum_{i=2}^n (g(z_i) - g(z_{i-1}))^2 + \frac{1}{n-1} \sum_{i=2}^n (e_i - e_{i-1})^2 \\ &\quad - \frac{2}{n-1} \sum_{i=2}^n (g(z_i) - g(z_{i-1})) (e_i - e_{i-1}). \end{aligned} \quad (5)$$

The logic behind the estimator (5) is that we hope terms with $g(z_i) - g(z_{i-1})$ will disappear due to the smoothness of the function (that asymptotically the difference will converge to zero).

Since z_i compose a uniform grid and are in increasing order, we know that $z_i - z_{i-1} = \frac{1}{n-1}$, and we can find the limit of the first term using the Lipschitz condition as

$$\begin{aligned} \left| \frac{1}{n-1} \sum_{i=2}^n (g(z_i) - g(z_{i-1}))^2 \right| &\leq \frac{G^2}{n-1} \sum_{i=2}^n (z_i - z_{i-1})^2 \\ &= \frac{G^2}{n-1} \frac{n-1}{(n-1)^2} = \frac{G^2}{(n-1)^2} \rightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$. We can similarly use the Lipschitz condition for the third term,

$$\begin{aligned} \left| \frac{2}{n-1} \sum_{i=2}^n (g(z_i) - g(z_{i-1})) (e_i - e_{i-1}) \right| &\leq \frac{2G}{n-1} \sum_{i=2}^n |z_i - z_{i-1}| |e_i - e_{i-1}| \\ &= \frac{2G}{(n-1)^2} \sum_{i=2}^n |e_i - e_{i-1}| \\ &\leq \frac{2G}{(n-1)^2} \sum_{i=2}^n (|e_i| + |e_{i+1}|) \\ &= \frac{2G}{n-1} \frac{1}{n-1} \sum_{i=2}^n (|e_i| + |e_{i-1}|) \xrightarrow{p} \frac{2G}{n-1} \cdot 2\mathbb{E}[|e_i|] \rightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$. The first inequality uses the Lipschitz condition of the function, the second inequality uses the triangular inequality, and the probability limit is followed by the law of large numbers and i.i.d. data. *Note that $\mathbb{E}[|e_i|] \neq 0$!* We assume that $\mathbb{E}[e_i|z_i] = 0$, but the absolute value takes into account only positive values of the random variable.

The second term results in the variance we are after:

$$\frac{1}{n-1} \sum_{i=2}^n (e_i - e_{i-1})^2 = \frac{1}{n-1} \sum_{i=2}^n e_i^2 - 2e_i e_{i-1} + e_{i-1}^2 \xrightarrow{p} 2\mathbb{E}[e_i^2] = 2\sigma.$$

Hence, the consistent estimator of the variance is

$$\hat{\sigma}^2 = \frac{1}{2} \frac{1}{n-1} \sum_{i=2}^n (y_i - y_{i-1})^2.$$

2. From the first-difference regression we have

$$\Delta y_i = \Delta x_i' \beta + \Delta g(z_i) + \Delta e_i$$

where Δ denotes the difference parameter. Consider an estimator of β ,

$$\hat{\beta} = \left(\sum_{i=2}^n \Delta x_i \Delta x_i' \right)^{-1} \sum_{i=2}^n \Delta x_i \Delta y_i. \quad (6)$$

Logic behind (6) is very similar to the one in the previous point: we take the difference and hope that the difference between $g(\cdot)$ function at neighboring points will converge to zero.

To prove consistency formally, we have

$$\hat{\beta} - \beta = \left(\frac{1}{n-1} \sum_{i=2}^n \Delta x_i \Delta x'_i \right)^{-1} \left(\frac{1}{n-1} \sum_{i=2}^n \Delta x_i (\Delta g(z_i) + \Delta e_i) \right).$$

The denominator is equal in the limit to some non-zero constant C by the law of large numbers,

$$\frac{1}{n-1} \sum_{i=2}^n \Delta x_i \Delta x'_i \xrightarrow{p} C.$$

The product

$$\frac{1}{n-1} \sum_{i=2}^n \Delta x_i \Delta e_i \xrightarrow{p} 0$$

due to the law of iterated expectations and $\mathbb{E}[e_i | x_i, z_i] = 0$. The remaining term,

$$\begin{aligned} \left| \frac{1}{n-1} \sum_{i=2}^n \Delta x_i \Delta g(z_i) \right| &= \left| \frac{1}{n-1} \sum_{i=2}^n \Delta x_i (g(z_i) - g(z_{i-1})) \right| \\ &\leq \frac{G}{n-1} \sum_{i=2}^n |\Delta x_i| |z_i - z_{i-1}| \\ &= \frac{G}{n-1} \frac{1}{n-1} \sum_{i=2}^n |\Delta x_i| \xrightarrow{p} 0 \end{aligned}$$

as $n \rightarrow 0$, where we again use the Lipschitz condition. As a result, $\hat{\beta} - \beta \xrightarrow{p} 0$.

Having found a consistent estimator of β , we can apply standard nonparametrics to the following regression:

$$y_i - x'_i \hat{\beta} = g(z_i) + e_i^*, \quad e_i^* := e_i + x'_i(\beta - \hat{\beta}).$$

Note that if $\hat{\beta}$ was not consistent, we would be working with a misspecified model since $\mathbb{E}[e_i^* | x_i, z_i] \neq 0$.

Consider the following estimator (we select the rectangular kernel for simplicity),

$$\begin{aligned} \hat{g}(z) &= \frac{\sum_{i=1}^n (y_i - x'_i \hat{\beta}) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^n \mathbb{1}\{|z_i - z| \leq h\}} \\ &= \frac{\sum_{i=1}^n (g(z_i) + e_i + x'_i(\beta - \hat{\beta}) \mathbb{1}\{|z_i - z| \leq h\})}{\sum_{i=1}^n \mathbb{1}\{|z_i - z| \leq h\}} \\ &= \frac{\sum_{i=1}^n g(z_i) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^n \mathbb{1}\{|z_i - z| \leq h\}} + \frac{\sum_{i=1}^n e_i \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^n \mathbb{1}\{|z_i - z| \leq h\}} + \frac{\sum_{i=1}^n x'_i(\beta - \hat{\beta}) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^n \mathbb{1}\{|z_i - z| \leq h\}} \end{aligned}$$

We work with the first term now. Notice that

$$\left| \frac{\sum_{i=1}^n (g(z_i) - g(z)) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^n \mathbb{1}\{|z_i - z| \leq h\}} \right| \leq G \left| \frac{\sum_{i=1}^n |z_i - z| \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^n \mathbb{1}\{|z_i - z| \leq h\}} \right| \leq Gh \quad (7)$$

which holds because all differences $|z_i - z|$ are less than h due to the indicator function, so we bound the whole expression by h . From it follows that

$$\frac{\sum_{i=1}^n g(z_i) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^n \mathbb{1}\{|z_i - z| \leq h\}} = \frac{\sum_{i=1}^n (g(z) + g(z_i) - g(z)) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^n \mathbb{1}\{|z_i - z| \leq h\}} \leq g(z) + Gh \rightarrow g(z)$$

as $h \rightarrow 0$ where we use (7).

For the third term, it holds by the law of iterated expectations and of large numbers that

$$\frac{\sum_{i=1}^n x'_i \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^n \mathbb{1}\{|z_i - z| \leq h\}} \xrightarrow{p} \frac{\mathbb{E}[x'_i \mathbb{1}\{|z_i - z| \leq h\}]}{\mathbb{E}[\mathbb{1}\{|z_i - z| \leq h\}]} = \mathbb{E}[x'_i],$$

and by $\hat{\beta} - \beta \xrightarrow{p} 0$, we have that

$$\frac{\sum_{i=1}^n x'_i (\beta - \hat{\beta}) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^n \mathbb{1}\{|z_i - z| \leq h\}} \xrightarrow{p} 0.$$

Finally, for the third term, we have by the law of iterated expectations

$$\frac{\sum_{i=1}^n e_i \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^n \mathbb{1}\{|z_i - z| \leq h\}} \xrightarrow{p} 0.$$

Hence, $\hat{g}(z) \xrightarrow{p} g(z)$.

Exercise 1.8. Let $(x_i, y_i)_{i=1}^n$ be a random sample from the population of (x, y) , where x has a discrete distribution with the support $a_{(1)}, \dots, a_{(k)}$, where $a_{(1)} < \dots < a_{(k)}$. Having written the conditional expectation $\mathbb{E}[y|x = a_{(j)}]$ in the form that allows to apply the analogy principle, propose an analog estimator $\hat{g}_j$ of $g_j = \mathbb{E}[y|x = a_{(j)}]$ and derive its asymptotic distribution.

Solution. Fix $a_{(j)}$ for $j = 1, \dots, k$. Then

$$g_j := g_i(a_{(j)}) = \mathbb{E}[y_i | x_i = a_{(j)}] = \frac{\mathbb{E}[y_i \mathbb{1}\{x_i = a_{(j)}\}]}{\mathbb{E}[\mathbb{1}\{x_i = a_{(j)}\}]},$$

which holds because

$$\begin{aligned} \mathbb{E}[\mathbb{1}\{x_i = a_{(j)}\}] &= \mathbb{E}[\mathbb{1}\{x_i = a_{(j)}\} | x_i = a_{(j)}] \cdot \mathbb{P}\{x_i = a_{(j)}\} \\ &\quad + \mathbb{E}[\mathbb{1}\{x_i = a_{(j)}\} | x_i \neq a_{(j)}] \cdot \mathbb{P}\{x_i \neq a_{(j)}\} \\ &= \mathbb{P}\{x_i = a_{(j)}\}, \end{aligned}$$

and

$$\begin{aligned} \mathbb{E}[y_i \mathbb{1}\{x_i = a_{(j)}\}] &= \mathbb{E}[y_i \mathbb{1}\{x_i = a_{(j)}\} | x_i = a_{(j)}] \cdot \mathbb{P}\{x_i = a_{(j)}\} \\ &\quad + \mathbb{E}[y_i \mathbb{1}\{x_i = a_{(j)}\} | x_i \neq a_{(j)}] \cdot \mathbb{P}\{x_i \neq a_{(j)}\} \\ &= \mathbb{E}[y_i | x_i = a_{(j)}] \cdot \mathbb{P}\{x_i = a_{(j)}\}. \end{aligned}$$

Hence, the estimator is

$$\hat{g}_i(a_{(j)}) = \frac{\sum_{i=1}^n y_i \mathbb{1}\{x_i = a_{(j)}\}}{\sum_{i=1}^n \mathbb{1}\{x_i = a_{(j)}\}} \xrightarrow{p} \frac{\mathbb{E}[y_i \mathbb{1}\{x_i = a_{(j)}\}]}{\mathbb{E}[\mathbb{1}\{x_i = a_{(j)}\}]} := g_i(a_{(j)}).$$

To derive the asymptotic distribution, normalize

$$\begin{aligned}\sqrt{n}(\hat{g}_j - g_j) &= \sqrt{n} \left(\frac{\sum_{i=1}^n y_i \mathbb{1}\{x_i = a_{(j)}\}}{\sum_{i=1}^n \mathbb{1}\{x_i = a_{(j)}\}} - g_j \right) \\ &= \sqrt{n} \left(\frac{\sum_{i=1}^n y_i \mathbb{1}\{x_i = a_{(j)}\} - g_j \sum_{i=1}^n \mathbb{1}\{x_i = a_{(j)}\}}{\sum_{i=1}^n \mathbb{1}\{x_i = a_{(j)}\}} \right) \\ &= \sqrt{n} \left(\frac{\sum_{i=1}^n (y_i - \mathbb{E}[y_i | x_i = a_{(j)}]) \mathbb{1}\{x_i = a_{(j)}\}}{\sum_{i=1}^n \mathbb{1}\{x_i = a_{(j)}\}} \right).\end{aligned}$$

So by the CLT, the numerator is distributed as

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n (y_i - \mathbb{E}[y_i | x_i = a_{(j)}]) \mathbb{1}\{x_i = a_{(j)}\} \xrightarrow{d} \mathcal{N}(0, \omega),$$

where ω is the asymptotic variance given by

$$\begin{aligned}\omega &:= \text{var} [(y_i - \mathbb{E}[y_i | x_i = a_{(j)}]) \mathbb{1}\{x_i = a_{(j)}\}] \\ &= \mathbb{E}[(y_i - \mathbb{E}[y_i | x_i = a_{(j)}])^2 | x_i = a_{(j)}] \cdot \mathbb{P}\{x_i = a_{(j)}\} \\ &= \text{var} [y_i | x_i = a_{(j)}] \cdot \mathbb{P}\{x_i = a_{(j)}\}.\end{aligned}$$

By the Slutsky's theorem and CMT, we have that

$$\sqrt{n}(\hat{g}_j - g_j) \xrightarrow{d} \mathcal{N}\left(0, \frac{\text{var} [y_i | x_i = a_{(j)}]}{\mathbb{P}\{x_i = a_{(j)}\}}\right).$$

2 Discrete choice models

Exercise 2.1. Let X be distributed as $\Lambda(x) := \frac{e^x}{1 + e^x}$. Verify that

- a) $\frac{\partial \Lambda(x)}{\partial x} = \Lambda(x)(1 - \Lambda(x))$,
- b) $\frac{\partial \log \Lambda(x)}{\partial x} = 1 - \Lambda(x)$,
- c) $-\frac{\partial^2 \log \Lambda(x)}{\partial x^2} = \Lambda(x)(1 - \Lambda(x))$.

Solution.

- a) Using the usual rule from calculus,

$$\frac{\partial \Lambda(x)}{\partial x} = \frac{e^x(1 + e^x) - e^x e^x}{(1 + e^x)^2} = \frac{e^x}{1 + e^x} \cdot \frac{1}{1 + e^x} = \Lambda(x)(1 - \Lambda(x)).$$

- b) First, $\log \Lambda(x) = \log\left(\frac{e^x}{1 + e^x}\right) = x - \log(1 + e^x)$, then

$$\frac{\partial \log \Lambda(x)}{\partial x} = 1 - \frac{e^x}{1 + e^x} = 1 - \Lambda(x).$$

- c) The result immediately follows from a) and b).

Lemma 2.2. For some constant a , we have

$$\int_{-\infty}^{\infty} a e^{-\varepsilon} \exp(-a e^{-\varepsilon}) d\varepsilon = 1.$$

Proof. Substitute $u = a e^{-\varepsilon}$. Then $\varepsilon = -\log u + \log a$, and $d\varepsilon = -(1/u)du$. We change the bounds of the integral as $\varepsilon \rightarrow -\infty \Rightarrow u \rightarrow \infty$, and $\varepsilon \rightarrow \infty \Rightarrow u \rightarrow 0$. Hence

$$-\int_{\infty}^0 u \exp(-u) \cdot \frac{1}{u} du = -\int_{\infty}^0 \exp(-u) du = \int_0^{\infty} \exp(-u) du = 1.$$

This completes the proof. □

Exercise 2.3. Derive the logit from the Gumbel distribution of ε_0 and ε_1 .

Solution. We start with

$$\begin{aligned}
\mathbb{P}\{y = 1\} &= \mathbb{P}\{\varepsilon_0 < \varepsilon_1 + V_1 - V_0\} \\
&= F_{\varepsilon_0}(\varepsilon_1 + V_1 - V_0) \\
&= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\varepsilon_1 + V_1 - V_0} f(\varepsilon_0, \varepsilon_1) d\varepsilon_0 \right) d\varepsilon_1 \\
&= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\varepsilon_1 + V_1 - V_0} f(\varepsilon_0) f(\varepsilon_1) d\varepsilon_0 \right) d\varepsilon_1
\end{aligned}$$

due to independence of ε_0 and ε_1 , so

$$\mathbb{P}\{y = 1\} = \int_{-\infty}^{\infty} f(\varepsilon_1) \left(\int_{-\infty}^{\varepsilon_1 + V_1 - V_0} f(\varepsilon_0) d\varepsilon_0 \right) d\varepsilon_1. \quad (8)$$

By our assumption,

$$f(\varepsilon_i) = e^{-\varepsilon_i} \exp(-e^{-\varepsilon_i}), \quad i = 1, 2.$$

Plugging it in (10), we have

$$\begin{aligned}
\mathbb{P}\{y = 1\} &= \int_{-\infty}^{\infty} f(\varepsilon_1) \left(\int_{-\infty}^{\varepsilon_1 + V_1 - V_0} e^{-\varepsilon_0} \exp(-e^{-\varepsilon_0}) d\varepsilon_0 \right) d\varepsilon_1 \\
&= \int_{-\infty}^{\infty} f(\varepsilon_1) [\exp(-e^{-\varepsilon_0})]_{-\infty}^{\varepsilon_1 + V_1 - V_0} d\varepsilon_1 \\
&= \int_{-\infty}^{\infty} f(\varepsilon_1) \exp(-e^{-(\varepsilon_1 + V_1 - V_0)}) d\varepsilon_1.
\end{aligned}$$

Now, plug in the density for $f(\varepsilon_1)$ to obtain

$$\begin{aligned}
\mathbb{P}\{y = 1\} &= \int_{-\infty}^{\infty} e^{-\varepsilon_1} \exp(-e^{-\varepsilon_1}) \exp(-e^{-(\varepsilon_1 + V_1 - V_0)}) d\varepsilon_1 \\
&= \int_{-\infty}^{\infty} e^{-\varepsilon_1} \exp\left((-e^{-\varepsilon_1} - e^{-(\varepsilon_1 + V_1 - V_0)})\right) d\varepsilon_1 \\
&= \int_{-\infty}^{\infty} e^{-\varepsilon_1} \exp\left((-e^{-\varepsilon_1} - e^{-\varepsilon_1} e^{-(V_1 - V_0)})\right) d\varepsilon_1 \\
&= \int_{-\infty}^{\infty} e^{-\varepsilon_1} \exp\left(-e^{-\varepsilon_1} (1 + e^{-(V_1 - V_0)})\right) d\varepsilon_1.
\end{aligned}$$

This integral is equal to

$$\mathbb{P}\{y = 1\} = \frac{1}{1 + e^{-(V_1 - V_0)}} = \frac{e^{V_1}}{e^{V_0} + e^{V_1}} = \frac{e^{V_1 - V_0}}{1 + e^{V_1 - V_0}} = \Lambda(V_1 - V_0)$$

due to the lemma above with $a := 1 + e^{-(V_1 - V_0)}$.

Exercise 2.4. Consider a latent variable modeled by $y_i^* = x_i' \beta + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, 1)$. Suppose we observe only $y_i = 1$ if $y_i^* < U_i$, and $y_i = 0$ if $y_i^* \geq U_i$, where the upper limit U_i is a known constant for each i and may differ over i .

- Find $\mathbb{P}\{y_i = 1 | x_i\}$.
- Provide details on how to estimate β with maximum likelihood.

Solution.

- Similarly as in the lecture,

$$\begin{aligned}
\mathbb{P}\{y_i = 1\} &= \mathbb{P}\{y_i^* < U_i\} \\
&= \mathbb{P}\{x_i' \beta + \varepsilon_i < U_i\} \\
&= \Phi(U_i - x_i' \beta).
\end{aligned}$$

b) Conditional likelihood is $\mathcal{L} = p^y(1-p)^{1-y}$ with $p = \Phi(U_i - x'_i\beta)$. Then,

$$\log \mathcal{L} = \sum_{i=1}^n y_i \log \Phi(U_i - x'_i\beta) + (1 - y_i) \log (1 - \Phi(U_i - x'_i\beta)).$$

Taking the first-order condition w.r.t. β ,

$$\begin{aligned} \frac{\partial \log \mathcal{L}}{\partial \beta} &= \sum_{i=1}^n -\frac{y_i}{\Phi(U_i - x'_i\beta)} \phi(U_i - x'_i\beta) x'_i + \frac{1 - y_i}{1 - \Phi(U_i - x'_i\beta)} \phi(U_i - x'_i\beta) x'_i \\ &= \sum_{i=1}^n \frac{\Phi(U_i - x'_i\beta) - y_i}{\Phi(U_i - x'_i\beta)(1 - \Phi(U_i - x'_i\beta))} \phi(U_i - x'_i\beta) x'_i = 0. \end{aligned}$$

Solving for β gives us the ML estimator of β .

Exercise 2.5. In 2022, CERGE-EI conducted a small research aimed at studying a relationship between drinking alcohol and the pass rate of the Econometrics II course.¹ Table 1 presents the number of students in each category. Dummy variable d_i takes a value of 1 if a student i has been drinking one drink per day during the final week, 0 otherwise.

Table 1: Data on 100 students

$d_i \setminus y_i$	fail	pass
no beer	24	28
beer	32	16

To answer this research question, CERGE-EI committee decided to specify the model as

$$y_i^* = \alpha + \beta d_i + \varepsilon_i, \quad i = 1, \dots, 100,$$

and they observe only

$$y_i = \begin{cases} 1 & \text{if } y_i^* > 0, \\ 0 & \text{otherwise.} \end{cases}$$

Obtain the maximum-likelihood estimators of α and β specifying the model as logit. Should you drink during the final week?

Solution. First, we write the log-likelihood function as

$$\begin{aligned} \log \mathcal{L}(\alpha, \beta) &= \sum_{0,0} \log \Pr\{y_i = 0, d_i = 0\} + \sum_{0,1} \log \Pr\{y_i = 0, d_i = 1\} \\ &\quad + \sum_{1,0} \log \Pr\{y_i = 1, d_i = 0\} + \sum_{1,1} \log \Pr\{y_i = 1, d_i = 1\} \end{aligned}$$

where $\sum_{i,j}$ is a summation of i that have $y_i = i$ and $d_i = j$. We calculate probabilities using the model assumptions:

- $\Pr\{y_i = 0, d_i = 0\} = \Pr\{\alpha + \varepsilon \leq 0\} = F(-\alpha) = 1 - F(\alpha),$
- $\Pr\{y_i = 0, d_i = 1\} = \Pr\{\alpha + \beta + \varepsilon \leq 0\} = F(-\alpha - \beta) = 1 - F(\alpha + \beta),$
- $\Pr\{y_i = 1, d_i = 0\} = \Pr\{\alpha + \varepsilon > 0\} = 1 - F(-\alpha) = F(\alpha),$
- $\Pr\{y_i = 1, d_i = 1\} = \Pr\{\alpha + \beta + \varepsilon > 0\} = 1 - F(-\alpha - \beta) = F(\alpha + \beta).$

Now we simplify the log-likelihood as follows:

$$\log \mathcal{L}(\alpha, \beta) = 24 \log(1 - F(\alpha)) + 32 \log(1 - F(\alpha + \beta)) + 28 \log F(\alpha) + 16 \log F(\alpha + \beta).$$

¹Real-life correspondence is incidental.

Taking the first-order condition w.r.t. α ,

$$\frac{\partial \log \mathcal{L}(\alpha, \beta)}{\partial \alpha} = -\frac{24}{1 - F(\alpha)} F'(\alpha) - \frac{32}{1 - F(\alpha + \beta)} F'(\alpha + \beta) + \frac{28}{F(\alpha)} F'(\alpha) + \frac{16}{F(\alpha + \beta)} F'(\alpha + \beta) = 0,$$

and w.r.t. β ,

$$\frac{\partial \log \mathcal{L}(\alpha, \beta)}{\partial \beta} = -\frac{32}{1 - F(\alpha + \beta)} F'(\alpha + \beta) + \frac{16}{F(\alpha + \beta)} F'(\alpha + \beta) = 0.$$

Substituting the second equation into the first one, the first one reduces to

$$\frac{24}{1 - F(\alpha)} F'(\alpha) = \frac{28}{F(\alpha)} F'(\alpha),$$

from which follows that $F(\alpha) = 28/52$.

Analogously, from the second equation,

$$\frac{32}{1 - F(\alpha + \beta)} F'(\alpha + \beta) = \frac{16}{F(\alpha + \beta)} F'(\alpha + \beta),$$

from which follows that $F(\alpha + \beta) = 16/48$.

Now, substitute for $F(x) = \frac{e^x}{1 + e^x}$ to have

$$F(\alpha) = \frac{e^{\hat{\alpha}}}{1 + e^{\hat{\alpha}}} = \frac{28}{52}.$$

By simple calculations, $\hat{\alpha} = 0.156$. Similarly,

$$F(\alpha + \beta) = \frac{e^{\hat{\alpha} + \hat{\beta}}}{1 + e^{\hat{\alpha} + \hat{\beta}}} = \frac{16}{48}.$$

By simple calculations, $\hat{\beta} = -0.850$. There is a clear negative effect of drinking on the pass rate.

Exercise 2.6. For the conditional logit,

$$p_{ij} = \frac{e^{x'_{ij}\beta}}{\sum_{\ell} e^{x'_{i\ell}\beta}},$$

show that

- a) $\frac{\partial p_{ij}}{\partial \beta} = p_{ij}(x_{ij} - \bar{x}_i)$, where $\bar{x}_i := \sum_{\ell=1}^m p_{i\ell} x_{i\ell}$, and
- b) $\frac{\partial p_{ij}}{\partial x_{ik}} = p_{ij}(\delta_{ijk} - p_{ik})\beta$, where $\delta_{ijk} := \mathbf{1}\{j = k\}$.

Solution.

- a) Differentiating by parts,

$$\begin{aligned} \frac{\partial p_{ij}}{\partial \beta} &= \frac{e^{x'_{ij}\beta} \cdot x_{ij} \cdot \sum_{\ell} e^{x'_{i\ell}\beta} - e^{x'_{ij}\beta} \cdot \sum_{\ell} e^{x'_{i\ell}\beta} \cdot x_{i\ell}}{(\sum_{\ell} e^{x'_{i\ell}\beta})^2} \\ &= \frac{e^{x'_{ij}\beta}}{\sum_{\ell} e^{x'_{i\ell}\beta}} x_{ij} - \frac{e^{x'_{ij}\beta}}{(\sum_{\ell} e^{x'_{i\ell}\beta})^2} \sum_{\ell} e^{x'_{i\ell}\beta} \cdot x_{i\ell} \\ &= p_{ij} x_{ij} - p_{ij} \sum_{\ell} p_{i\ell} x_{i\ell} = p_{ij}(x_{ij} - \bar{x}_i). \end{aligned}$$

- b) For the marginal effect, we differentiate between $j = k$ and $j \neq k$. For the first case,

$$\frac{\partial p_{ij}}{\partial x_{ij}} = \frac{e^{x'_{ij}\beta} \cdot \beta}{\sum_{\ell} e^{x'_{i\ell}\beta}} - \frac{e^{x'_{ij}\beta}}{(\sum_{\ell} e^{x'_{i\ell}\beta})^2} e^{x'_{ij}\beta} \cdot \beta = p_{ij}(1 - p_{ij})\beta,$$

and for the second case,

$$\frac{\partial p_{ij}}{\partial x_{ik}} = -\frac{e^{x'_{ij}\beta}}{(\sum_{\ell} e^{x'_{i\ell}\beta})^2} \cdot e^{x'_{ik}\beta} \cdot \beta = -p_{ij}p_{ik}\beta,$$

so altogether,

$$\frac{\partial p_{ij}}{\partial x_{ij}} = p_{ij}(\delta_{ijk} - p_{ik})\beta.$$

Exercise 2.7. You have 20 observations that are drawn from a censored normal distribution such that 6 are equal to zero. The applicable model is

$$y_i^* = \mu + \varepsilon_i, \quad y_i = \begin{cases} y_i^* & \text{if } \mu + \varepsilon_i > 0 \\ 0 & \text{otherwise} \end{cases}, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

- Discuss the OLS estimator $\hat{\mu}$ of μ . Do you expect this estimator to over- or underestimate μ ?
- If we consider only the nonzero observations, then the truncated regression model applies. Discuss the OLS estimator $\tilde{\mu}$ of μ in this case. Do you expect this estimator to over- or underestimate μ ? Is $\hat{\mu} > \tilde{\mu}$?
- Consider the tobit model that applies to the full data set. Formulate the log-likelihood. Reformulate the log-likelihood in terms of $\theta := 1/\sigma$ and $\gamma := \mu/\sigma$. Derive the necessary conditions for maximizing the log-likelihood with respect to θ and γ .
- Repeat c) in the context of the truncated regression model.

Solution.

- The OLS estimator in this case is just a sample mean. Because all negative values have been transformed into zeroes, the estimator $\hat{\mu}$ would overestimate the true mean.
- Likewise, for the truncated mean $\tilde{\mu}$, whereas a complete sample might include some negative values, the observed one does not. Once again, this will serve to inflate the estimator of the mean. Because the censored mean includes zeros, we have $\tilde{\mu} > \hat{\mu}$ (see the following Figure 1 where censored (green) and truncated (blue) means are computed for 1000 iterations with $\mu = 2$).

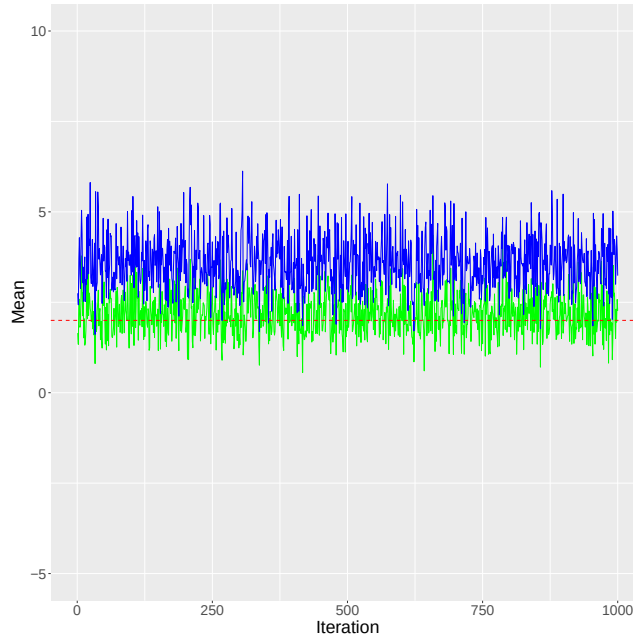


Figure 1: Censored and truncated mean

- c) Denote the number of observations which are non-zero as n_1 , and the number of zero (censored) observations as n_0 . Then, $n = n_0 + n_1$. The log-likelihood for the tobit model is then

$$\log \mathcal{L}(y_i; \mu, \sigma^2) = -\frac{n_1}{2} (\log 2\pi + \log \sigma^2) - \frac{1}{2\sigma^2} \sum_{n_1} (y_i - \mu)^2 + \sum_{n_0} \log \left(1 - \Phi \left(\frac{\mu}{\sigma} \right) \right),$$

and with $\theta := 1/\sigma$ and $\gamma := \mu/\sigma$,

$$\log \mathcal{L}(y_i; \theta, \gamma) = -\frac{n_1}{2} \left(\log 2\pi + \log \frac{1}{\theta^2} \right) - \frac{1}{2} \sum_{n_1} (\theta y_i - \gamma)^2 + \sum_{n_0} \log (1 - \Phi(\gamma)).$$

The first-order conditions are

$$\begin{aligned} \frac{\partial \mathcal{L}(y_i; \theta, \gamma)}{\partial \gamma} &= \sum_{n_1} (\theta y_i - \gamma) - \sum_{n_0} \frac{\phi(\gamma)}{1 - \Phi(\gamma)} \\ &= \sum_{n_1} \theta y_i - n_1 \gamma - n_0 \frac{\phi(\gamma)}{1 - \Phi(\gamma)} = 0, \end{aligned}$$

and

$$\frac{\partial \mathcal{L}(y_i; \theta, \gamma)}{\partial \theta} = \frac{n_1}{\theta} - \sum_{n_1} (\theta y_i - \gamma) y_i = 0.$$

- d) In the truncated case, the log-likelihood for the tobit model is

$$\log \mathcal{L}(y_i; \mu, \sigma^2) = -\frac{n_1}{2} (\log 2\pi + \log \sigma^2) - \frac{1}{2\sigma^2} \sum_{n_1} (y_i - \mu)^2 - \sum_{n_1} \log \Phi \left(\frac{\mu}{\sigma} \right),$$

or transformed,

$$\log \mathcal{L}(y_i; \theta, \gamma) = -\frac{n_1}{2} \left(\log 2\pi + \log \frac{1}{\theta^2} \right) - \frac{1}{2} \sum_{n_1} (\theta y_i - \gamma)^2 - \sum_{n_1} \log \Phi(\gamma).$$

Taking the first-order conditions,

$$\frac{\partial \log \mathcal{L}(y_i; \theta, \gamma)}{\partial \gamma} = \sum_{n_1} (\theta y_i - \gamma) - n_1 \frac{\phi(\gamma)}{\Phi(\gamma)} = 0,$$

and

$$\frac{\partial \log \mathcal{L}(y_i; \theta, \gamma)}{\partial \theta} = \frac{n}{\theta} - \sum_{n_1} (\theta y_i - \gamma) y_i = 0.$$

Exercise 2.8. We have data for $n = 753$ married women aged 30-60 in the US in 1975. 428 women in the sample work, and 325 do not work.

- a) For the sample of working women, the following equation was estimated in *Stata* by the OLS:

$$\log(\text{wage})_i = \beta_0 + \beta_1 \text{age}_i + \beta_2 \text{educ}_i + \beta_3 \text{exper}_i + \beta_4 \text{expersq}_i + u_i$$

with the following results.

Is the OLS estimator biased?

- b) Suppose that you have the following variables that can be used as exclusion restrictions for the Heckman selection correction:

- **kids05**: the number of kids between 0 and 5,
- **kids618**: the number of kids between 6 and 18,
- **nwifeinc**: household income excluding wife's income.

Describe how would you implement it.

- c) The Heckman selection correction was implemented using **kids05**, **kids618**, and **nwifeinc** as exclusion restrictions with the following results:

Comment on the sign of the parameter for the inverse Mills ratio (**imr**). Is there evidence for the selection bias?

Linear regression				Number of obs	=	428
				F(4, 423)	=	20.42
				Prob > F	=	0.0000
				R-squared	=	0.1568
				Root MSE	=	.6672
lwage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	.0002836	.0054379	0.05	0.958	-.0104051	.0109724
educ	.1075228	.0133046	8.08	0.000	.0813715	.133674
exper	.0415623	.0153042	2.72	0.007	.0114805	.0716442
expersq	-.0008152	.0004135	-1.97	0.049	-.001628	-2.33e-06
_cons	-.5333761	.2876625	-1.85	0.064	-1.098802	.0320499

Figure 2: OLS results for a)

Linear regression				Number of obs	=	428
				F(5, 422)	=	17.71
				Prob > F	=	0.0000
				R-squared	=	0.1570
				Root MSE	=	.66794
lwage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	-.0006676	.0071394	-0.09	0.926	-.0147009	.0133657
educ	.1095345	.0154197	7.10	0.000	.0792255	.1398434
exper	.0447027	.0183525	2.44	0.015	.0086291	.0807763
expersq	-.0008664	.0004269	-2.03	0.043	-.0017055	-.0000272
imr	.0434592	.2127619	0.20	0.838	-.374746	.4616643
_cons	-.5708801	.3076202	-1.86	0.064	-1.175539	.0337785

Figure 3: Heckman selection correction for c)

Solution.

- The OLS will be biased due to the self-selection bias.
- The first step is to run a probit on the probability of working (using the entire sample) with original regressors and the exclusion restrictions. Based on the probit estimates, compute the inverse Mills ratio as

$$\text{IMR} = \frac{\phi(x'\hat{\beta})}{\Phi(x'\hat{\beta})}.$$

Then, include this as a regressor in the OLS.

- Coefficient on the inverse Mills ratio is positive, $\hat{\lambda} = 0.04 > 0$, so
 - women with higher unobserved propensity to work also have a higher unobserved productivity at work,
 - working women are a positive selection with respect to wages,
 - positive correlation of error terms in the selection and outcome equation.

But $\hat{\lambda}$ is not significant, which means that there is no significant evidence for the sample selection bias. In fact, the OLS and Heckman coefficient are very similar.

3 Panel data

Exercise 3.1. Consider the model with individual effects,

$$y_{it} = \alpha_i + x'_{it}\beta + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T$$

Heckman selection model -- two-step estimates (regression model with sample selection)				Number of obs	=	753
				Selected	=	428
				Nonselected	=	325
				Wald chi2(4)	=	51.52
				Prob > chi2	=	0.0000
lwage	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
lwage						
age	-.0006676	.0060686	-0.11	0.912	-.0125619	.0112267
educ	.1095345	.0161004	6.80	0.000	.0779782	.1410907
exper	.0447027	.0178719	2.50	0.012	.0096743	.079731
expersq	-.0008664	.0004439	-1.95	0.051	-.0017364	3.70e-06
_cons	-.5708801	.3120663	-1.83	0.067	-1.182519	.0407586
select						
age	-.0528527	.0084772	-6.23	0.000	-.0694678	-.0362376
educ	.1309047	.0252542	5.18	0.000	.0814074	.180402
exper	.1233476	.0187164	6.59	0.000	.0866641	.1600311
expersq	-.0018871	.0006	-3.15	0.002	-.003063	-.0007111
kids05	-.8683285	.1185223	-7.33	0.000	-1.100628	-.636029
kids618	.036005	.0434768	0.83	0.408	-.049208	.1212179
nwifeinc	-.0120237	.0048398	-2.48	0.013	-.0215096	-.0025378
_cons	.2700768	.508593	0.53	0.595	-.7267473	1.266901
/mills						
lambda	.0434592	.1679665	0.26	0.796	-.2857492	.3726675

Figure 4: Heckman selection correction: correct se

with $\beta \in \mathbb{R}^{k \times 1}$.

- Write down the formula for the within estimator of β .
- Indicate cases for which not all elements of β could be estimated.
- Assume that all elements of β could be estimated and specify the formula for the within estimator when $T = 2$. Show that this leads to a regression which could be written as

$$\begin{pmatrix} y_1^* \\ -y_1^* \end{pmatrix} = \begin{pmatrix} X_1^* \\ -X_1^* \end{pmatrix} \beta + \text{error}.$$

Specify y_1^* and X_1^* in terms of y_{it} and x_{it} .

- Show that the regression in c) yields an estimator that is equivalent to the first-differences estimator.

Exercise 3.2. Modify the random effects model by the addition of a time specific disturbance. So that

$$y_{it} = u_i + v_t + x'_{it}\beta + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T$$

where we assume that

- $\mathbb{E}[\varepsilon_{it}] = \mathbb{E}[u_i] = \mathbb{E}[v_t] = 0$,
- $\mathbb{E}[\varepsilon_{it}u_j] = \mathbb{E}[\varepsilon_{it}v_s] = \mathbb{E}[u_iv_t] = 0$, for $\forall i, j, t, s$,
- $\text{var}[\varepsilon_{it}] = \sigma_\varepsilon^2$, $\text{cov}[\varepsilon_{it}, \varepsilon_{js}] = 0$, for $\forall i, j, t, s$,
- $\text{var}[u_i] = \sigma_u^2$, $\text{cov}[u_i, u_j] = 0$, for $\forall i, j$,
- $\text{var}[v_t] = \sigma_v^2$, $\text{cov}[v_t, v_s] = 0$, for $\forall t, s$.

Write out the full covariance matrix for a data set with $N = T = 2$.

Solution. Let $\mu_{it} := \varepsilon_{it} + u_i + v_t$. Let's first derive the diagonal elements of the variance covariance matrix,

$$\begin{aligned}\mathbb{E}[\mu_{it}^2] &= \mathbb{E}[(\varepsilon_{it} + u_i + v_t)^2] \\ &= \mathbb{E}[\varepsilon_{it}^2] + \mathbb{E}[u_i^2] + \mathbb{E}[v_t^2] + 2\mathbb{E}[\varepsilon_{it}u_i] + \mathbb{E}[\varepsilon_{it}v_t] + \mathbb{E}[v_tu_i] \\ &= \mathbb{E}[\varepsilon_{it}^2] + \mathbb{E}[u_i^2] + \mathbb{E}[v_t^2] \\ &= \sigma_\varepsilon^2 + \sigma_u^2 + \sigma_v^2,\end{aligned}$$

where we use the fact that the covariance between all three terms are zero.

Now, for a given individual in different time period, the non-diagonal elements of the variance covariance matrix are

$$\begin{aligned}\mathbb{E}[\mu_{it}\mu_{is}] &= \mathbb{E}[(\varepsilon_{it} + u_i + v_t)(\varepsilon_{is} + u_i + v_s)] \\ &= \mathbb{E}[\varepsilon_{it}\varepsilon_{is}] + \mathbb{E}[u_i^2] + \mathbb{E}[v_tv_s] \\ &\quad + \mathbb{E}[\varepsilon_{it}u_i] + \mathbb{E}[\varepsilon_{is}u_i] + \mathbb{E}[\varepsilon_{is}v_t] + \mathbb{E}[\varepsilon_{it}v_s] + \mathbb{E}[v_tu_i] + \mathbb{E}[v_su_i] \\ &= \mathbb{E}[u_i^2] \\ &= \sigma_u^2,\end{aligned}$$

where not only all covariances between the elements are zero, but also the serial correlations of ε_{it} and v_t .

Next, for a given period and different individuals, non-diagonal elements of the variance covariance matrix are

$$\begin{aligned}\mathbb{E}[\mu_{it}\mu_{jt}] &= \mathbb{E}[(\varepsilon_{it} + u_i + v_t)(\varepsilon_{jt} + u_j + v_t)] \\ &= \mathbb{E}[\varepsilon_{it}\varepsilon_{jt}] + \mathbb{E}[u_iu_j] + \mathbb{E}[v_t^2] + \text{cov}[\dots] \\ &= \mathbb{E}[v_t^2] \\ &= \sigma_v^2,\end{aligned}$$

where not only all covariances between the elements are zero, but also ε_{it} and u_i are uncorrelated across individuals by assumption.

Lastly, if both $s \neq t$ and $i \neq j$, the terms are zero. Thus, when $N = T = 2$, the variance covariance matrix is

$$\begin{pmatrix} \sigma_\varepsilon^2 + \sigma_u^2 + \sigma_v^2 & \sigma_u^2 & \sigma_v^2 & 0 \\ \sigma_u^2 & \sigma_\varepsilon^2 + \sigma_u^2 + \sigma_v^2 & 0 & \sigma_v^2 \\ \sigma_v^2 & 0 & \sigma_\varepsilon^2 + \sigma_u^2 + \sigma_v^2 & \sigma_u^2 \\ 0 & \sigma_v^2 & \sigma_u^2 & \sigma_\varepsilon^2 + \sigma_u^2 + \sigma_v^2 \end{pmatrix}.$$

Exercise 3.3. The two-way fixed effects model is

$$y_{it} = u_i + v_t + x'_{it}\beta + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T$$

where u_i and v_t are the individual-specific and the time-specific intercepts.

- Show that the within estimator of β , which is the best linear unbiased, could be obtained by applying two within (one-way) transformations of the model.
- Show that the order of these two withing (one-way) transformations is unimportant. Give an intuitive explanation for this result.

Solution.

- Consider the following transformations of y_{it} (and similarly of other variables):

$$\bar{y}_i := \frac{1}{T} \sum_{t=1}^T y_{it}, \quad \bar{y}_t := \frac{1}{N} \sum_{i=1}^N y_{it}, \quad \bar{y} := \frac{1}{NT} \sum_{t=1}^T \sum_{i=1}^N y_{it},$$

where $\bar{y}_i$ is averaging over time, $\bar{y}_t$ is averaging over individuals, and $\bar{y}$ is averaging over time and individuals.

Then, the first within transformation is

$$y_{it} - \bar{y}_t = (x_{it} - \bar{x}_t)' \beta + \varepsilon_{it} - \bar{\varepsilon}_t + u_i - \bar{u}_t,$$

and the second transformation,

$$(y_{it} - \bar{y}_t) - \frac{1}{T} \sum_{t=1}^T (y_{it} - \bar{y}_t) = \left[(x_{it} - \bar{x}_t) - \frac{1}{T} \sum_{t=1}^T (x_{it} - \bar{x}_t) \right]' \beta + (\varepsilon_{it} - \bar{\varepsilon}_t) - \frac{1}{T} \sum_{t=1}^T (\varepsilon_{it} - \bar{\varepsilon}_t),$$

where we have that $u_i - \bar{u}_t - \frac{1}{T} \sum_{t=1}^T (u_i - \bar{u}_t) = u_i - \bar{u} - (u_i - \bar{u}) = 0$. So, it can be also written as

$$y_{it} - \bar{y}_t - \bar{y}_i + \bar{y} = (x_{it} - \bar{x}_t - \bar{x}_i + \bar{x})' \beta + \varepsilon_{it} - \bar{\varepsilon}_t - \bar{\varepsilon}_i + \bar{\varepsilon}.$$

- b) It can be seen directly from the equation that the order of the averaging does not matter for the final outcome (both averages are subtracted). Note that it is equivalent when the panel data is balanced, i.e. each i is observed for T_i periods and $T_i = T$ for all i .

Exercise 3.4. Verify that the infeasible GLS estimator simplifies to the OLS estimator of

$$y_{it}^* = x_{it}^{*\prime} \beta + \varepsilon_{it}^*, \quad i = 1, \dots, N, \quad t = 1, \dots, T$$

where

$$y_{it}^* := y_{it} - (1 - \theta)\bar{y}_i, \quad x_{it}^* := x_{it} - (1 - \theta)\bar{x}_i, \quad \varepsilon_{it}^* := \theta\alpha_i + \varepsilon_{it} - (1 - \theta)\bar{\varepsilon}_i,$$

and

$$\theta^2 := \frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + T\sigma_\alpha^2}.$$

Solution. Consider the model

$$y_{it} = \alpha_i + x_{it}' \beta + \varepsilon_{it},$$

and a stacked across time version of it,

$$y_i = e\alpha_i + x_i' \beta + \varepsilon_i$$

where $y_i \in \mathbb{R}^{T \times 1}$, $e := (1, \dots, 1)' \in \mathbb{R}^{T \times 1}$, $x_i \in \mathbb{R}^{T \times k}$, and $\varepsilon_i \in \mathbb{R}^{T \times 1}$.

To use the GLS, we have to find a variance covariance matrix of the error term in this model (assume x_i to be fixed). So,

$$\begin{aligned} \Omega &:= \text{var} [e\alpha_i + x_i' \beta + \varepsilon_i] \\ &= \mathbb{E}[(e\alpha_i + x_i' \beta + \varepsilon_i)(e\alpha_i + x_i' \beta + \varepsilon_i)'] \\ &= \mathbb{E}[\varepsilon_i \varepsilon_i'] + \mathbb{E}[\alpha_i^2] e e' = \sigma_\varepsilon^2 I_T + \sigma_\alpha^2 e e', \end{aligned}$$

where I_T is a $T \times T$ identity matrix. Introducing $Q := I_T - T^{-1} e e'$, we can rewrite it as

$$\Omega = \sigma_\varepsilon^2 \left(Q + \frac{1}{\theta^2} (I_T - Q) \right).$$

Because $Q Q' = Q$, we have $\Omega^{-1} = \sigma_\varepsilon^{-2} (Q + \theta^2 (I_T - Q))$, and

$$\Omega^{-1/2} = \frac{1}{\sigma_\varepsilon} (Q + \theta (I_T - Q)). \tag{9}$$

Multiplying our model by (9) will give us the GLS estimator of β . For example, with y_i we have

$$\begin{aligned} (Q + \theta (I_T - Q)) y_i &= y_i - e \bar{y}_i' + \theta (y_i - (y_i - e \bar{y}_i')) \\ &= y_i - e \bar{y}_i' (1 - \theta). \end{aligned}$$

Performing similar algebra for x_i , $e\alpha_i$, and ε_i yields

$$y_i - (1 - \theta) e \bar{y}_i' = (x_i - (1 - \theta) \bar{x})' \beta + \theta e \alpha_i + \varepsilon_i - (1 - \theta) e \bar{\varepsilon}_i'.$$

Unstacking the equation (writing out explicitly for each T), leads to the required result.

Exercise 3.5. Suppose that the random effects model $y_{it} = x_{it}' \beta + \eta_i + v_{it}$ is to be estimated with a panel in which the groups have different numbers of observations. Let T_i be the number of observations in group i . Show that the pooled least squares estimator is unbiased and consistent despite this complication.

Solution. The model is equivalent to

$$y_i = X_i\beta + v_i + \eta_i\iota, \quad y_i \in \mathbb{R}^{T_i}, \quad X_i \in \mathbb{R}^{T_i \times K}, \quad v_i \in \mathbb{R}^{T_i}, \quad \iota := (1, \dots, 1)' \in \mathbb{R}^{T_i}, \quad i = 1, \dots, n,$$

and given the random effects model assumption, $\mathbb{E}[\eta_i x_{it}] = 0$. The pooled OLS estimator of β is

$$\hat{\beta} = \left(\sum_{i=1}^n X_i' X_i \right)^{-1} \sum_{i=1}^n X_i' y_i$$

given that $\sum_{i=1}^n X_i' X_i$ is invertible. To show the bias, rewrite

$$\begin{aligned} \hat{\beta} &= \left(\sum_{i=1}^n X_i' X_i \right)^{-1} \sum_{i=1}^n X_i' (X_i\beta + v_i + \eta_i\iota) \\ &= \beta + \left(\sum_{i=1}^n X_i' X_i \right)^{-1} \sum_{i=1}^n X_i' (v_i + \eta_i\iota) \\ &= \beta + \left(\sum_{i=1}^n X_i' X_i \right)^{-1} \sum_{i=1}^n X_i' u_i \end{aligned}$$

with $u_i := v_i + \eta_i\iota$. Hence, the bias $\mathbb{E}[\hat{\beta} - \beta | X_i]$ is zero if $\mathbb{E}[X_i' u_i | X_i] = X_i' \mathbb{E}[u_i | X_i] = X_i' (\mathbb{E}[v_i | X_i] + \mathbb{E}[\eta_i | X_i]) = 0$. It holds because the first expectation is zero by the i.i.d. independent mean zero errors v_{it} , and the second expectation is zero by the random effects assumption and the law of iterated expectations.

To show consistency, rewrite

$$\hat{\beta} - \beta = \left(\frac{1}{n} \sum_{i=1}^n X_i' X_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_i' u_i \right).$$

As $n \rightarrow \infty$, using the weak law of large numbers and Slutsky's theorem, we have that

$$\hat{\beta} - \beta \xrightarrow{P} Q^{-1} \mathbb{E}[X_i' u_i],$$

where $\lim_{n \rightarrow \infty} n^{-1} \sum_{i=1}^n X_i' X_i = \mathbb{E}[X_i' X_i] := Q$ is a non-deficient matrix with full rank. Under the random effects assumption and arguments as above, we have that $\mathbb{E}[X_i' u_i] = 0$. Hence, the estimator is consistent.

Exercise 3.6. Consider $y_{it} = x_{it}'\beta + \eta_i + v_{it}$, $i = 1, \dots, N$, $t = 1, \dots, T$, where $v_{it} \sim \mathcal{N}(0, \sigma^2)$ and $\beta = 0$. Write out the likelihood for estimating η_i and σ^2 , and show that the MLE estimator $\hat{\sigma}^2$ is biased when $T < \infty$.

Solution. From the setup, it implies that $y_{it} \sim \mathcal{N}(\eta_i, \sigma^2)$. The individual log-likelihood for each i (across T) is then

$$\log \mathcal{L}(y_{it} | \eta_i, \sigma^2) = C_0 - \frac{T}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{t=1}^T (y_{it} - \eta_i)^2$$

where C_0 is some constant independent of η_i and σ^2 . The ML estimator of η_i is the solution to the equation

$$\frac{\partial \log \mathcal{L}(y_{it} | \eta_i, \sigma^2)}{\partial \eta_i} = \sum_{t=1}^T (y_{it} - \hat{\eta}_i) = 0,$$

which is $\hat{\eta}_i = T^{-1} \sum_{t=1}^T y_{it} := \bar{y}_i$.

To estimate σ^2 , we use the joint log-likelihood (across i and T),

$$\log \mathcal{L}(y_{it} | \eta_i, \sigma^2) = C_1 - \frac{NT}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^N \sum_{t=1}^T (y_{it} - \eta_i)^2$$

where C_1 is some constant independent of η_i and σ^2 . The ML estimator of σ^2 is the solution to the equation

$$\frac{\partial \log \mathcal{L}(y_{it} | \eta_i, \sigma^2)}{\partial \sigma^2} = -\frac{NT}{2} \cdot \frac{1}{\hat{\sigma}^2} - \frac{1}{2\hat{\sigma}^4} \sum_{i=1}^N \sum_{t=1}^T (y_{it} - \hat{\eta}_i)^2 = 0.$$

Substituting for $\hat{\eta}_i$ and rearranging, we have

$$\hat{\sigma}^2 = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T (y_{it} - \bar{y}_i)^2.$$

Expectation of the estimator is

$$\begin{aligned} \mathbb{E}[\hat{\sigma}^2] &= \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T (y_{it} - \bar{y}_i)^2 \right] \\ &= \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T y_{it}^2 - \frac{2}{T} \sum_{t=1}^T y_{it} \bar{y}_i + \frac{1}{T} \sum_{t=1}^T \bar{y}_i^2 \right] \\ &= \sigma^2 - \frac{2}{T} \sum_{t=1}^T \mathbb{E}[y_{it} \bar{y}_i] + \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\bar{y}_i^2] \\ &= \sigma^2 - \frac{2}{T} \sigma^2 + \frac{1}{T} \sigma^2 = \sigma^2 - \frac{\sigma^2}{T}, \end{aligned}$$

which is not equal to σ^2 unless $T \rightarrow \infty$.

Exercise 3.7. Consider $y_{it} = \mathbf{1}\{x_{it}\beta + \eta_i + v_{it} \geq 0\}$, where the errors v_{it} have the logistic cdf. Consider $T = 2$, $x_{i1} = 0$ and $x_{i2} = 1$, and show that the sufficient statistic for η_i is $y_{i1} + y_{i2} = 1$, i.e. conditioning on $y_{i1} + y_{i2} = 1$ implies that the MLE does not depend on η_i .

Solution. The log-likelihood function for two periods is given by

$$\begin{aligned} \log \mathcal{L}(y_i | x_i, \beta, \eta_i) &= y_{i1} \log \Lambda(x_{i1}\beta + \eta_i) + (1 - y_{i1}) \log(1 - \Lambda(x_{i1}\beta + \eta_i)) \\ &\quad + y_{i2} \log \Lambda(x_{i2}\beta + \eta_i) + (1 - y_{i2}) \log(1 - \Lambda(x_{i2}\beta + \eta_i)), \end{aligned}$$

where $\Lambda(z) = 1/(1 + e^{-z})$, and given known values of the covariates,

$$\log \mathcal{L}(y_i | \beta, \eta_i) = y_{i1} \log \Lambda(\eta_i) + (1 - y_{i1}) \log(1 - \Lambda(\eta_i)) + y_{i2} \log \Lambda(\beta + \eta_i) + (1 - y_{i2}) \log(1 - \Lambda(\beta + \eta_i)).$$

Taking the first derivative w.r.t. η_i , and using $\Lambda(z)'/\Lambda(z) = (1 - \Lambda(z))$, we have

$$\frac{\partial \log \mathcal{L}(y_i | \beta, \eta_i)}{\partial \eta_i} = y_{i1}(1 - \Lambda(\hat{\eta}_i)) - (1 - y_{i1})\Lambda(\hat{\eta}_i) + y_{i2}(1 - \Lambda(\hat{\eta}_i + \beta)) - (1 - y_{i2})\Lambda(\hat{\eta}_i + \beta) = 0,$$

which implies

$$y_{i1} + y_{i2} = \Lambda(\hat{\eta}_i) + \Lambda(\hat{\eta}_i + \beta).$$

Now, we discuss three cases:

1. if $y_{i1} + y_{i2} = 0$, $\hat{\eta}_i = -\infty$,
2. if $y_{i1} + y_{i2} = 2$, $\hat{\eta}_i = \infty$,
3. if $y_{i1} + y_{i2} = 1$, $2\hat{\eta}_i = \beta$, and $\hat{\eta}_i = -\beta/2$.

Hence, in the case 3., it is possible to identify η_i from the estimate of β only. It implies that conditional on $\zeta_i := y_{i1} + y_{i2} = 1$, the log-likelihood is independent on η_i making ζ_i a sufficient statistic.

Exercise 3.8. Derive the bias of the OLS estimator for α in a dynamic panel of the form $y_{it} = \alpha y_{it-1} + \eta_i + v_{it}$. Are there any conditions on α that should hold for the estimator to be well-defined?

Solution. First, rewrite the model in recursive form,

$$\begin{aligned} y_{it} &= \alpha(\alpha y_{it-2} + \eta_i + v_{it-1}) + \eta_i + v_{it} \\ &= \alpha^2(\alpha y_{it-3} + \eta_i + v_{it-2}) + \alpha\eta_i + \alpha v_{it-1} + \eta_i + v_{it} \\ &= \dots \\ &= \alpha^t y_0 + \left(\sum_{s=0}^{t-1} \alpha^s \right) \eta_i + \sum_{s=0}^{t-1} \alpha^s v_{it-s}. \end{aligned}$$

The bias of the OLS estimator is driven by $\mathbb{E}[y_{it-1}u_{it}]$, where $u_{it} := \eta_i + v_{it}$. Using the property of the geometric series, $\sum_{k=0}^{n-1} r^k = \frac{1-r^n}{1-r}$ for $|r| < 1$, we have

$$\begin{aligned} y_{it-1} &= \alpha^{t-1}y_0 + \left(\sum_{s=0}^{t-2} \alpha^s\right) \eta_i + \sum_{s=0}^{t-2} \alpha^s v_{it-s} \\ &= \alpha^{t-1}y_0 + \left(\frac{1-\alpha^{t-1}}{1-\alpha}\right) \eta_i + \sum_{s=0}^{t-2} \alpha^s v_{it-s}. \end{aligned}$$

Now taking expectations,

$$\begin{aligned} \mathbb{E}[y_{it-1}\eta_i] &= \mathbb{E}\left[\alpha^{t-1}y_0\eta_i + \left(\frac{1-\alpha^{t-1}}{1-\alpha}\right) \eta_i^2 + \sum_{s=0}^{t-2} \alpha^s v_{it-s}\eta_i\right] \\ &= \alpha^{t-1}\mathbb{E}[y_0\eta_i] + \left(\frac{1-\alpha^{t-1}}{1-\alpha}\right) \sigma_\eta^2 \neq 0. \end{aligned}$$

Because we have that $\mathbb{E}[y_{it-1}u_{it}] = \mathbb{E}[y_{it-1}\eta_i] + \mathbb{E}[y_{it-1}v_{it}]$, the bias is nonzero even if $\mathbb{E}[y_{it-1}v_{it}] = 0$.

Exercise 3.9. Consider the panel AR(1) model with individual effects,

$$y_{it} = \alpha y_{it-1} + \eta_i + v_{it}$$

where $\eta_i \sim \text{i.i.d.}(0, \sigma_\eta^2)$ and $v_{it} \sim \text{i.i.d.}(0, \sigma_v^2)$ are mutually independent, and for all i we have $y_{i0} = 0$. Derive $\text{var}[y_{it}]$ for $t = 1, \dots, T$.

Solution. In the previous exercise we have shown that

$$y_{it} = \alpha^t y_0 + \left(\sum_{s=0}^{t-1} \alpha^s\right) \eta_i + \sum_{s=0}^{t-1} \alpha^s v_{it-s},$$

and given that $y_{i0} = 0$, we have

$$y_{it} = \left(\sum_{s=0}^{t-1} \alpha^s\right) \eta_i + \sum_{s=0}^{t-1} \alpha^s v_{it-s}.$$

Now, the variance is

$$\text{var}\left[\sum_{s=0}^{t-1} \alpha^s v_{it-s}\right] = \sigma_v^2 \sum_{s=0}^{t-1} \alpha^{2s} = \sigma_v^2 \frac{1-\alpha^{2t}}{1-\alpha^2},$$

so that

$$\text{var}[y_{it}] = \sigma_\eta^2 \left(\frac{1-\alpha^t}{1-\alpha}\right)^2 + \sigma_v^2 \frac{1-\alpha^{2t}}{1-\alpha^2}.$$

Exercise 3.10. Assume that we are in the AR(1) dynamic model setup such that

$$y_{it} = \alpha y_{it-1} + \eta_i + v_{it}$$

but now our v_{it} follows an MA(1) process such that

$$v_{it} = w_{it} + b w_{it-1},$$

where $w_{it} \sim \text{i.i.d.}(0, \sigma_w^2)$ (i.e. v_{it} is serially correlated). Show that in this case the instrument y_{it-2} is not a valid instrument for estimating α with GMM in first differences, while the instruments y_{it-j} for $j \geq 3$ remain valid.

Solution. Given that $v_{it} \sim \text{MA}(1)$, the autocovariance of order $s = 0$ is

$$\begin{aligned} \mathbb{E}[v_{it}v_{it}] &= \mathbb{E}[(w_{it} + b w_{it-1})(w_{it} + b w_{it-1})] \\ &= \mathbb{E}[w_{it}^2] + 2b\mathbb{E}[w_{it}w_{it-1}] + b^2\mathbb{E}[w_{it-1}^2] \\ &= \sigma_w^2 + b^2\sigma_w^2 = \sigma_w^2(1+b^2), \end{aligned}$$

of order $s = 1$ is

$$\begin{aligned}\mathbb{E}[v_{it}v_{it-1}] &= \mathbb{E}[(w_{it} + bw_{it-1})(w_{it-1} + bw_{it-2})] \\ &= \mathbb{E}[w_{it}w_{it-1}] + b\mathbb{E}[w_{it-1}^2] + b\mathbb{E}[w_{it}w_{it-2}] + b^2\mathbb{E}[w_{it-1}w_{it-2}] = b\sigma_w^2,\end{aligned}$$

and of order $s > 1$ is

$$\mathbb{E}[v_{it}v_{it-s}] = \mathbb{E}[(w_{it} + bw_{it-1})(w_{it-s} + bw_{it-s-1})] = \dots = 0,$$

where we use the fact that w_{it} and w_{it-s} for $\forall s$ is independent by assumption of i.i.d. data. The first differences model is then

$$y_{it} - y_{it-1} = \alpha(y_{it-1} - y_{it-2}) + v_{it} - v_{it-1},$$

so to be valid, the instrument should not be correlated with the error $u_{it} := v_{it} - v_{it-1}$. In other words, for y_{it-2} to be valid, we require

$$\mathbb{E}[u_{it}y_{it-2}] = 0,$$

which holds if v_{it-2} (that is inside y_{it-2}) is uncorrelated to v_{it} and v_{it-1} . However, we have shown that

$$\mathbb{E}[v_{it}v_{it-1}] = \mathbb{E}[v_{it-1}v_{it-2}] = b\sigma_w^2 \neq 0,$$

which in turns implies that y_{it-2} is not a valid instrument.

On the other hand, any instrument that is of the form y_{it-j} for $j > 2$ is valid, because

$$\mathbb{E}[y_{it-j}u_{it}] = 0$$

due to $\mathbb{E}[v_{it}v_{it-s}] = 0$ for $s > 1$ with $s = j - 1$.

Exercise 3.11. We have data for a panel of companies on gross investment expenditures I_{it} and net capital stock K_{it} . We model the investment rate $y_{it} = I_{it}/K_{it}$ as

$$\left(\frac{I_{it}}{K_{it}}\right) = \alpha \left(\frac{I_{it-1}}{K_{it-1}}\right) + \eta_i + v_{it},$$

and Table 3 shows the results of estimating the model in *levels* by OLS and WG, and the model in *first differences* with one instrument, two instruments, and all Arellano-Bond instruments. For the last two estimators, it also shows the Sargan test statistic and the m_2 statistic for second-order serial correlation in the residuals from the estimated model.

Table 2: Estimation results (703 firms, 4966 observations)

	OLS	WG	2SLS DIF	GMM DIF	GMM DIF
	(1)	(2)	(3)	(4)	(5)
$\hat{\alpha}$	0.2669 (.0185)	-0.0094 (.0181)	0.1626 (.0362)	0.1593 (.0327)	0.1560 (.0318)
m_2				0.52	0.46
Sargan test				0.36	0.43
Instruments			$(I/K)_{t-2}$	$(I/K)_{t-2}$ $(I/K)_{t-3}$	$(I/K)_{t-2}$ $(I/K)_{t-3}$ $\vdots$ $(I/K)_1$

- For each of the models in columns (2) and (3), write down the estimated equation(s).
- Comment on the estimates of α in each of the columns. Are the results in line with theory (in terms of possible bias of the different estimators)? Why do we need to use instruments?
- Comment on the standard errors of the last three estimators. Are the results in line with theory?
- For the two GMM estimators (column (4) and (5)), what do you conclude from the two specification tests? What are these tests' null hypotheses and why are these useful to run?

Solution.

- a) In column (2), the WG model is estimated, which takes away individual averages over time. Thus, the estimated equation is

$$\left(\frac{I_{it}}{K_{it}} - \frac{\bar{I}_i}{\bar{K}_i} \right) = \alpha \left(\frac{I_{it-1}}{K_{it-1}} - \frac{\bar{I}_i}{\bar{K}_i} \right) + v_{it} - \bar{v}_i,$$

where $\frac{\bar{I}_i}{\bar{K}_i} = T^{-1} \sum_{t=1}^T \frac{I_{it}}{K_{it}}$.

In column (3), the model is estimated in first differences,

$$\left(\frac{I_{it}}{K_{it}} - \frac{I_{it-1}}{K_{it-1}} \right) = \alpha \left(\frac{I_{it-1}}{K_{it-1}} - \frac{I_{it-2}}{K_{it-2}} \right) + v_{it} - v_{it-1}.$$

The estimation is done using 2SLS with the instrument $\frac{I_{it-2}}{K_{it-2}}$, so that in the first stage we project the endogenous regressor on the instrument,

$$\left(\frac{I_{it-1}}{K_{it-1}} - \frac{I_{it-2}}{K_{it-2}} \right) = \gamma_0 + \gamma_1 \frac{I_{it-2}}{K_{it-2}} + \varepsilon_{it},$$

and in the second stage we project the outcome variable on fitted values from the first stage,

$$\left(\frac{I_{it}}{K_{it}} - \frac{I_{it-1}}{K_{it-1}} \right) = \alpha \left(\widehat{\frac{I_{it-1}}{K_{it-1}} - \frac{I_{it-2}}{K_{it-2}}} \right) + v_{it} - v_{it-1}$$

where $\left(\widehat{\frac{I_{it-1}}{K_{it-1}} - \frac{I_{it-2}}{K_{it-2}}} \right)$ are fitted values, $\hat{\gamma}_1 \left(\frac{I_{it-2}}{K_{it-2}} \right)$.

- b) If the specified model is correct, and we have

$$\left(\frac{I_{it}}{K_{it}} \right) = \alpha \left(\frac{I_{it-1}}{K_{it-1}} \right) + \eta_i + v_{it},$$

we are in a dynamic panel setting and running OLS and WG will deliver biased estimates. Even if $\mathbb{E}[y_{it-1}v_{it}] = 0$, we have

$$\lim_{N \rightarrow \infty} \hat{\alpha}_{OLS} > \alpha, \quad \lim_{N \rightarrow \infty} \hat{\alpha}_{WG} < \alpha,$$

because $\mathbb{E}[y_{it-1}\eta_i] > 0$. Thus, OLS will overestimate α whereas WG will underestimate α . However, if we estimate the model in first differences and use instruments (as in columns (3), (4), and (5)) we will get consistent estimates of α under the assumption that these are valid instruments. The instrument used in (3) implies the Anderson-Hsiao estimator for α , column (4) expands the number of instruments, and column (5) uses all Arellano-Bond instruments. These rely on the moment conditions $\mathbb{E}[y_{t-j}\Delta v_{it}] = 0$ for $j \geq 2$ which are satisfied as long as v_{it} are serially uncorrelated.

We need instruments in the first place, because the regressor $\left(\frac{I_{it-1}}{K_{it-1}} - \frac{I_{it-2}}{K_{it-2}} \right)$ is by construction correlated with the error term $v_{it} - v_{it-1}$.

- c) The standard errors of the three last columns are decreasing which is what we expect from theory when adding valid instruments. The efficiency gains from the extra instruments are very small though.
- d) The two specification tests are the Sargan overidentification test and the test for second-order serial correlation in the residuals.

The Sargan overidentification test's null hypothesis is that the moment conditions are satisfied (the instruments are valid). The Sargan test statistic is distributed as a χ^2 with $(L - K)$ degrees of freedom (which are 1 and $T - 3$ in columns (4) and (5)) and in both models we obtain a low statistic of 0.36 and 0.43. Thus, we cannot reject the null hypothesis that the moment conditions are satisfied, and thus the instruments are valid. This test is crucial to determine whether we are using valid instruments and thus have a consistent estimate of α .

The test for second-order serial correlation in the residuals has as the null hypothesis that the residuals from the first-difference model are *not* second-order serially correlated. If this is the case then it implies that there is no correlation between $v_{it} - v_{it-1}$ and $v_{it-2} - v_{it-3}$ which would indicate that v_{it} are serially uncorrelated. This is a crucial assumption on which the validity of the instruments rests. The test statistic m_2 is normally distributed and with the reported m_2 of 0.52 and 0.46 we cannot reject the null hypothesis that the residuals from the first-difference model are *not* second-order serially correlated. Both specification tests go in the same direction and both imply that the instruments used are valid.

4 Causal inference

Exercise 4.1. In Table below, you have the potential outcomes of 4 cancer patients with their potential outcomes and treatment assignment for $D_i = 0$ corresponding to undergoing chemotherapy and $D_i = 1$ corresponding to surgery.

Table 3: Surgery vs therapy

patient	Y_i^1	Y_i^0	δ_i	Y_i	D_i
<i>case 1</i>					
1	4	3	1	4	1
1	6	5	1	6	1
3	1	2	-1	2	0
4	3	6	-3	6	0
<i>case 2</i>					
1	5	3	2	5	1
1	6	5	1	6	1
3	7	6	1	6	0
4	10	8	2	8	0

Show that in one case there is no selection bias, whereas in the other there is no heterogeneous treatment effects bias.

Solution. *Case 1.* Given that individual treatment effects are available, we can compute the average treatment effect (ATE) directly as

$$\text{ATE} = \frac{1 + 1 - 1 - 3}{4} = -0.5.$$

The naive approach of estimating it using the simple difference in outcomes (SDO) would yield

$$\text{SDO} = \frac{4 + 6}{2} - \frac{2 + 6}{2} = 1.$$

In general, it holds that $\text{SDO} = \text{ATE} + \text{SB} + \text{HTB}$, where SB is the selection bias and HTB is the heterogeneous treatment effect bias. Because we have that $\text{ATE} \neq \text{SDO}$, we conclude that one (or both) of the biases is nonzero.

The selection bias is given by

$$\text{SB} = \mathbb{E}[Y_i^0 | D_i = 1] - \mathbb{E}[Y_i^0 | D_i = 0] = \frac{3 + 5}{2} - \frac{2 + 6}{2} = 0.$$

For the heterogeneous treatment effect bias, we need to compute the average treatment effect on treated and untreated (ATT and ATU) respectively,

$$\text{ATT} = \mathbb{E}[\delta_i | D_i = 1] = \frac{1 + 1}{2} = 1, \quad \text{ATU} = \mathbb{E}[\delta_i | D_i = 0] = \frac{-1 - 3}{2} = -2.$$

The HTB is then

$$\text{HTB} = (1 - \pi)(\text{ATT} - \text{ATU}) = 0.5(1 + 2) = 1.5$$

with π denoting the proportion of the treated observations.

Case 2. For the second case, we briefly state the results:

$$\text{ATE} = \frac{2 + 1 + 1 + 2}{4} = 1.5,$$

$$\text{SDO} = \frac{5 + 6}{2} - \frac{6 + 8}{2} = -1.5,$$

$$\text{ATT} = \frac{2 + 1}{2} = 1.5,$$

$$\text{ATU} = \frac{1 + 2}{2} = 1.5,$$

$$\text{SB} = \frac{3 + 5}{2} - \frac{6 + 8}{2} = -3,$$

$$\text{HTB} = 0.$$

Exercise 4.2. Show that running a regression

$$y_i = \beta_0 + \beta_1 D_i + \varepsilon_i, \quad i = 1, \dots, N, \quad \mathbb{E}[\varepsilon_i | D_i] = 0,$$

leads to the OLS estimate of β_1 given by

$$\hat{\beta}_1 = \frac{1}{N_1} \sum_{i=1}^{N_1} (y_i | D_i = 1) - \frac{1}{N_0} \sum_{i=1}^{N_0} (y_i | D_i = 0),$$

which is a simple difference in means.

Solution. For subsequent derivations, we define $\bar{y}_1 := \frac{1}{N_1} \sum_{i=1}^{N_1} (y_i | D_i = 1)$ and $\bar{y}_0 := \frac{1}{N_0} \sum_{i=1}^{N_0} (y_i | D_i = 0)$ so that we have to show $\hat{\beta}_1 = \bar{y}_1 - \bar{y}_0$. The OLS estimator is given by

$$\hat{\beta}_1 = \frac{\sum_{i=1}^N (D_i - \bar{D})(y_i - \bar{y})}{\sum_{i=1}^N (D_i - \bar{D})^2},$$

where $\bar{D} = N^{-1} \sum_{i=1}^N D_i = N_1/N$, and $\bar{y} = N^{-1} \sum_{i=1}^N y_i$. Expanding the numerator, we have

$$\sum_{i=1}^N (D_i - \bar{D})(y_i - \bar{y}) = \sum_{i=1}^N D_i y_i - D_i \bar{y} - \bar{D} y_i + \bar{D} \bar{y}.$$

Working with each term separately,

- $\sum_{i=1}^N D_i y_i = \frac{N_1}{N_1} \sum_{i=1}^N D_i y_i = N_1 \cdot \frac{1}{N_1} \sum_{i=1, D_i=1} y_i = N_1 \cdot \bar{y}_1,$
- $-\sum_{i=1}^N \bar{D} y_i = -\sum_{i=1}^N \frac{N_1}{N} y_i = -N_1 \cdot \bar{y},$
- $-\sum_{i=1}^N D_i \bar{y} = -\sum_{i=1, D_i=1} \bar{y} = -N_1 \cdot \bar{y},$
- $\sum_{i=1}^N \bar{D} \bar{y} = \sum_{i=1}^N \frac{N_1}{N} \bar{y} = N_1 \cdot \bar{y},$

we have that

$$\sum_{i=1}^N (D_i - \bar{D})(y_i - \bar{y}) = N_1(\bar{y}_1 - \bar{y}) = \left(N_1 - \frac{N_1^2}{N} \right) (\bar{y}_1 - \bar{y}_0)$$

where we substitute for $\bar{y}$ using the fact that $N = N_0 + N_1$, and $\bar{y} = N^{-1}(N_1 \bar{y}_1 + N_0 \bar{y}_0)$.

Now, for the denominator, we have

$$\begin{aligned} \sum_{i=1}^N (D_i - \bar{D})^2 &= \sum_{i=1}^N D_i^2 - 2D_i \bar{D} + \bar{D}^2 \\ &= \sum_{i=1}^N D_i^2 - 2 \sum_{i=1}^N D_i \bar{D} + \sum_{i=1}^N \bar{D}^2 \\ &= N_1 - 2 \sum_{i=1}^N D_i \frac{N_1}{N} + \sum_{i=1}^N \frac{N_1^2}{N^2} \\ &= N_1 - \frac{N_1^2}{N}. \end{aligned}$$

Putting everything together, we have

$$\hat{\beta}_1 = \frac{\left(N_1 - \frac{N_1^2}{N} \right) (\bar{y}_1 - \bar{y}_0)}{N_1 - \frac{N_1^2}{N}} = \bar{y}_1 - \bar{y}_0.$$

Exercise 4.3. Show that running a regression

$$y_i = \beta_0 + \beta_1 D_i + \varepsilon_i, \quad i = 1, \dots, N, \quad \mathbb{E}[\varepsilon_i | D_i] = 0,$$

with heteroskedasticity-robust standard errors provides an estimate that is numerically equivalent to the variance of the average treatment effect computed directly.

Solution. In the previous exercise we showed that $\hat{\beta}_1 = \bar{y}_1 - \bar{y}_0$. Answering the question in the setup is equivalent to showing that

$$\widehat{\text{var}}[\hat{\beta}_1] = \widehat{\text{var}}[\bar{y}_1] - \widehat{\text{var}}[\bar{y}_0] = \hat{\sigma}_1^2/N_1 + \hat{\sigma}_0^2/N_0 \equiv \widehat{\text{var}}_{\text{EHW}}[\hat{\beta}_1]$$

where $\hat{\sigma}_k^2$, $k \in \{0, 1\}$ are estimated variances of the outcome computed on treated and untreated respectively, and $\widehat{\text{var}}_{\text{EHW}}[\hat{\beta}_1]$ are standard Eicker-Huber-White (EHW) standard errors.

Denote $x_i := (1 \ D_i)'$. Then the asymptotic heteroskedasticity-robust variance of the OLS estimator is

$$\text{var} \left[\begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} \right] = \frac{1}{N} \mathbb{E}[x_i x_i']^{-1} \mathbb{E}[x_i x_i' \varepsilon_i^2] \mathbb{E}[x_i x_i']^{-1}.$$

Denoting residuals as $\hat{\varepsilon}_i := y_i - x_i' \hat{\beta}$, the estimator is then the usual EHW estimator,

$$\begin{aligned} \widehat{\text{var}} \left[\begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} \right] &= \frac{1}{N} \left(\frac{1}{N} \sum_{i=1}^N x_i x_i' \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N x_i x_i' \hat{\varepsilon}_i^2 \right) \left(\frac{1}{N} \sum_{i=1}^N x_i x_i' \right)^{-1} \\ &= \frac{1}{N} \left(\frac{1}{N} \sum_{i=1}^N \begin{pmatrix} 1 & D_i \\ D_i & D_i \end{pmatrix} \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \begin{pmatrix} 1 & D_i \\ D_i & D_i \end{pmatrix} \hat{\varepsilon}_i^2 \right) \left(\frac{1}{N} \sum_{i=1}^N \begin{pmatrix} 1 & D_i \\ D_i & D_i \end{pmatrix} \right)^{-1} \\ &= \frac{1}{N} \begin{pmatrix} 1 & N_1/N \\ N_1/N & N_1/N \end{pmatrix}^{-1} \begin{pmatrix} N^{-1} \sum_{i=1}^N \hat{\varepsilon}_i^2 & N^{-1} \sum_{i=1, D_i=1} \hat{\varepsilon}_i^2 \\ N^{-1} \sum_{i=1, D_i=1} \hat{\varepsilon}_i^2 & N^{-1} \sum_{i=1, D_i=1} \hat{\varepsilon}_i^2 \end{pmatrix} \begin{pmatrix} 1 & N_1/N \\ N_1/N & N_1/N \end{pmatrix}^{-1} \\ &= \begin{pmatrix} N & N_1 \\ N_1 & N_1 \end{pmatrix}^{-1} \begin{pmatrix} \sum_{i=1}^N \hat{\varepsilon}_i^2 & \sum_{i=1, D_i=1} \hat{\varepsilon}_i^2 \\ \sum_{i=1, D_i=1} \hat{\varepsilon}_i^2 & \sum_{i=1, D_i=1} \hat{\varepsilon}_i^2 \end{pmatrix} \begin{pmatrix} N & N_1 \\ N_1 & N_1 \end{pmatrix}^{-1}. \end{aligned}$$

Computing the inverses, and using the fact that $N = N_0 + N_1$, and $\sum_{i=1}^N \hat{\varepsilon}_i^2 = \sum_{i=1, D_i=0} \hat{\varepsilon}_i^2 + \sum_{i=1, D_i=1} \hat{\varepsilon}_i^2$ we have

$$\begin{aligned} \widehat{\text{var}} \left[\begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} \right] &= \begin{pmatrix} 1/N_0 & -1/N_0 \\ -1/N_0 & 1/N_0 + 1/N_1 \end{pmatrix} \begin{pmatrix} \sum_{i=1}^N \hat{\varepsilon}_i^2 & \sum_{i=1, D_i=1} \hat{\varepsilon}_i^2 \\ \sum_{i=1, D_i=1} \hat{\varepsilon}_i^2 & \sum_{i=1, D_i=1} \hat{\varepsilon}_i^2 \end{pmatrix} \begin{pmatrix} 1/N_0 & -1/N_0 \\ -1/N_0 & 1/N_0 + 1/N_1 \end{pmatrix} \\ &= \begin{pmatrix} N_0^{-1} \sum_{i=1, D_i=0} \hat{\varepsilon}_i^2 & 0 \\ N_1^{-1} \sum_{i=1, D_i=1} \hat{\varepsilon}_i^2 - N_0^{-1} \sum_{i=1, D_i=0} \hat{\varepsilon}_i^2 & N_1^{-1} \sum_{i=1, D_i=1} \hat{\varepsilon}_i^2 \end{pmatrix} \begin{pmatrix} 1/N_0 & -1/N_0 \\ -1/N_0 & 1/N_0 + 1/N_1 \end{pmatrix} \\ &= \begin{pmatrix} N_0^{-2} \sum_{i=1, D_i=0} \hat{\varepsilon}_i^2 & -N_0^{-2} \sum_{i=1, D_i=0} \hat{\varepsilon}_i^2 \\ -N_0^{-2} \sum_{i=1, D_i=0} \hat{\varepsilon}_i^2 & N_0^{-2} \sum_{i=1, D_i=0} \hat{\varepsilon}_i^2 + N_1^{-2} \sum_{i=1, D_i=1} \hat{\varepsilon}_i^2 \end{pmatrix}. \end{aligned}$$

We note that the low-right element is exactly the estimated variance of $\hat{\beta}_1$,

$$\begin{aligned} \widehat{\text{var}}[\hat{\beta}_1] &= N_0^{-2} \sum_{i=1, D_i=0} \hat{\varepsilon}_i^2 + N_1^{-2} \sum_{i=1, D_i=1} \hat{\varepsilon}_i^2 \\ &:= \hat{\sigma}_0^2/N_0 + \hat{\sigma}_1^2/N_1. \end{aligned}$$

Exercise 4.4. Show that *i*) if $p(X)$ is the propensity score, then $X \perp D \mid p(X)$; *ii*) if $Y^1, Y^0 \perp D \mid X$, then assignment to treatment is unconfounded given the propensity score, i.e. $Y^1, Y^0 \perp D \mid p(X)$.

Solution. For *i*), first, note that

$$\mathbb{P}\{D = 1 \mid X, p(X)\} = \mathbb{E}[D \mid X, p(X)] = \mathbb{E}[D \mid X] = \mathbb{P}\{D = 1 \mid X\} := p(X),$$

and

$$\mathbb{P}\{D = 1 \mid p(X)\} = \mathbb{E}[D \mid p(X)] = \mathbb{E}[D \mid p(X), X] = \mathbb{E}[\mathbb{E}[D \mid p(X), X] \mid p(X)] = \mathbb{E}[p(X) \mid p(X)] = p(X).$$

So in total we have that

$$\mathbb{P}\{D = 1 \mid X, p(X)\} = \mathbb{P}\{D = 1 \mid p(X)\},$$

that is, conditional on $p(X)$, probability of receiving a treatment is the same with or without conditioning on the observed covariates X . In other words, conditional on $p(X)$, the treatment is independent of covariates.

For *ii*), we have that

$$\begin{aligned}
\mathbb{P}\{D = 1|Y^1, Y^0, p(X)\} &= \mathbb{E}[D|Y^1, Y^0, p(X)] \\
&= \mathbb{E}_X [\mathbb{E}[D|X, Y^1, Y^0]|Y^1, Y^0, p(X)] \\
&= \mathbb{E}_X [\mathbb{E}[D|X]|Y^1, Y^0, p(X)] \\
&= \mathbb{E}_X [p(X)|Y^1, Y^0, p(X)] = p(X),
\end{aligned}$$

where the third equality follows from the "if" assumption in the setup. From *i*), we have that

$$\mathbb{P}\{D = 1|p(X)\} = \mathbb{P}\{D = 1|Y^1, Y^0, p(X)\} = p(X),$$

that is, conditional on $p(X)$, the treatment and the potential outcomes are independent.

Exercise 4.5. Show that the IV coefficient given by

$$\delta_{IV} = \frac{\text{cov}[Z_i, Y_i]}{\text{cov}[Z_i, D_i]}$$

results in

$$\delta_W = \frac{\mathbb{E}[Y_i|Z_i = 1] - \mathbb{E}[Y_i|Z_i = 0]}{\mathbb{E}[D_i|Z_i = 1] - \mathbb{E}[D_i|Z_i = 0]}.$$

Solution. First, the numerator of δ_{IV} is

$$\begin{aligned}
\text{cov}[Z_i, Y_i] &= \mathbb{E}[Z_i Y_i] - \mathbb{E}[Z_i]\mathbb{E}[Y_i] \\
&= \mathbb{E}[Y_i|Z_i = 1] \cdot \mathbb{P}\{Z_i = 1\} - \mathbb{P}\{Z_i = 1\} \cdot (\mathbb{E}[Y_i|Z_i = 1] \cdot \mathbb{P}\{Z_i = 1\} + \mathbb{E}[Y_i|Z_i = 0] \cdot \mathbb{P}\{Z_i = 0\}) \\
&= \mathbb{E}[Y_i|Z_i = 1] \cdot \mathbb{P}\{Z_i = 1\} - \mathbb{P}\{Z_i = 1\}^2 \mathbb{E}[Y_i|Z_i = 1] - \mathbb{P}\{Z_i = 1\}(1 - \mathbb{P}\{Z_i = 1\})\mathbb{E}[Y_i|Z_i = 0] \\
&= (\mathbb{E}[Y_i|Z_i = 1] - \mathbb{E}[Y_i|Z_i = 0]) \mathbb{P}\{Z_i = 1\}(1 - \mathbb{P}\{Z_i = 1\}).
\end{aligned}$$

Next, the denominator is by analogy is

$$\text{cov}[Z_i, D_i] = (\mathbb{E}[D_i|Z_i = 1] - \mathbb{E}[D_i|Z_i = 0]) \mathbb{P}\{Z_i = 1\}(1 - \mathbb{P}\{Z_i = 1\}),$$

so that division $\text{cov}[Z_i, Y_i]/\text{cov}[Z_i, D_i]$ yields δ_W .

Exercise 4.6. Suppose that with heterogeneous treatment effects instead of monotonicity, we assume an eligibility rule of the form

$$\mathbb{P}(D_i = 1|Z_i = 0) = 0.$$

Show that in this case, we can identify the average treatment effect on the treated.

Solution. First, compute

$$\begin{aligned}
\mathbb{E}[Y_i|Z_i = 1] &= \mathbb{E}[Y_i^0] + \mathbb{E}[(Y_i^1 - Y_i^0)D_i|Z_i = 1] \\
&= \mathbb{E}[Y_i^0] + \mathbb{E}[\mathbb{E}[(Y_i^1 - Y_i^0)D_i|D_i = 1, Z_i = 1]|Z_i = 1] \\
&= \mathbb{E}[Y_i^0] + \mathbb{E}[Y_i^1 - Y_i^0|D_i = 1, Z_i = 1] \cdot \mathbb{P}\{D_i = 1|Z_i = 1\} \\
&= \mathbb{E}[Y_i^0] + \mathbb{E}[Y_i^1 - Y_i^0|D_i = 1] \cdot \mathbb{P}\{D_i = 1|Z_i = 1\},
\end{aligned}$$

where the last line follows from the eligibility rule assumption, because the event $D_i = 1$ is sufficient for the event $Z_i = 1$.

Similarly,

$$\begin{aligned}
\mathbb{E}[Y_i|Z_i = 0] &= \mathbb{E}[Y_i^0] + \mathbb{E}[(Y_i^1 - Y_i^0)D_i|Z_i = 0] \\
&= \mathbb{E}[Y_i^0] + \mathbb{E}[(Y_i^1 - Y_i^0)D_i = 1, Z_i = 0] \cdot \mathbb{P}(D_i = 1|Z_i = 0) \\
&= \mathbb{E}[Y_i^0]
\end{aligned}$$

due to the eligibility rule assumption. It follows that

$$\frac{\mathbb{E}[Y_i|Z_i = 1] - \mathbb{E}[Y_i|Z_i = 0]}{\mathbb{P}(D_i = 1|Z_i = 1)} = \mathbb{E}[Y_i^1 - Y_i^0|D_i = 1].$$

Year	Draft eligibility ^a	Number of deaths ^b	Number of suicides ^c	Probability of death ^d	Probability of suicide	Probability of military service ^e
1950	Yes	2,601	436	.0204 (.0004)	.0034 (.0002)	.3527 (.0325)
	No	2,169	352	.0195 (.0004)	.0032 (.0002)	.1934 (.0233)
<i>Difference (Yes minus No)</i>				.0009 (.0006)	.0002 (.0002)	.1593 (.0401)
<i>IV estimates^f</i>				.0056 (.0040)	.0013 (.0013)	
1951	Yes	1,494	279	.0170 (.0004)	.0032 (.0002)	.2831 (.0390)
	No	2,823	480	.0168 (.0003)	.0029 (.0001)	.1468 (.0180)
<i>Difference (Yes minus No)</i>				.0002 (.0005)	.0003 (.0002)	.1362 (.0429)
<i>IV estimates</i>				.0015 (.0037)	.0022 (.0016)	

Figure 5: Data on civilian mortality (Angrist, Imbens, and Rubin, 1996)

By definition, $ATT := \mathbb{E}[Y_i^1 - Y_i^0 | D_i = 1]$ so we have that

$$ATT = \frac{\mathbb{E}[Y_i | Z_i = 1] - \mathbb{E}[Y_i | Z_i = 0]}{\mathbb{P}(D_i = 1 | Z_i = 1)} = \frac{\mathbb{E}[Y_i | Z_i = 1] - \mathbb{E}[Y_i | Z_i = 0]}{\mathbb{E}[D_i | Z_i = 1]},$$

which is the Wald estimand having $\mathbb{E}[D_i | Z_i = 0] = 0$ by the eligibility rule assumption.

Exercise 4.7. Table 5 contains data on civilian mortality for white men born in 1950 and 1951. For the men born in 1951, interpret the reduced form (intention-to-treat) and the IV estimates.

Solution. What are the intention-to-treat (ITT) effects here? Remember that ITT effects come from the reduced form regression, in which we regress the outcome variable (here, probability of death or suicide) on the assignment to treatment (drafting to the army). The coefficient of the regression where there is only the constant and the assignment to treatment,

$$Y_i = \alpha + \gamma Z_i + \varepsilon_i,$$

is equivalent to the mean comparison between the two groups,

$$\gamma = \mathbb{E}[Y_i | Z_i = 1] - \mathbb{E}[Y_i | Z_i = 0].$$

Thus, the *Difference (Yes minus No)* line tells us what is the effect of the assignment on the probability of death and suicide. Men born in 1951 who were assigned treatment had a 0.02% higher probability to die compared to those not assigned, and 0.03% higher probability to commit suicide. These are only ITT effects as we know there were men that were treated and not treated in both assignment groups.

The IV results are based on the Wald estimand,

$$\delta_W = \frac{\mathbb{E}[Y_i | Z_i = 1] - \mathbb{E}[Y_i | Z_i = 0]}{\mathbb{E}[D_i | Z_i = 1] - \mathbb{E}[D_i | Z_i = 0]}.$$

Note that the Wald estimand is essentially dividing the ITT by the difference in the probability of serving for those assigned and not assigned (which is shown in the last column). In fact, the IV results are

$$\hat{\delta}_{W, \text{death}} = \frac{0.0002}{0.1362} = 0.0015, \quad \hat{\delta}_{W, \text{suicide}} = \frac{0.0003}{0.1362} = 0.0022.$$

We know that the Wald estimand gives the LATE, that is, the average causal effect for the compliers (those who were induced to serve by the draft). Note, however, that most effects are not statistically significant.

Exercise 4.8. You have seen in class that with sharp regression discontinuity design (RDD), it is important to correctly specify the underlying model of how the continuous variable impacts the outcome variable. One possible misspecification shown was when a nonlinear model with no discontinuity is fit by a linear model.

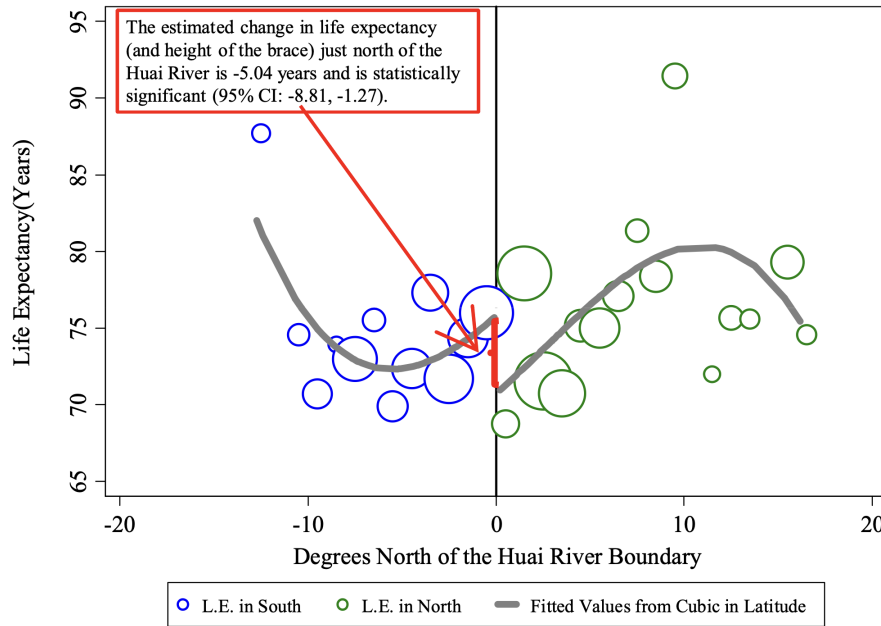


Figure 6: Estimated change in life expectancy (Chen, Ebenstein, Greenstone, and Li, 2013)

In the paper “*Evidence on the impact of sustained exposure to air pollution on life expectancy from China’s Huai river policy*” by Chen, Ebenstein, Greenstone, and Li, the authors use China’s Huai river policy as a sharp discontinuity to estimate the effect of air pollution on life expectancy. China’s Huai river policy provided free winter heating via provision of coal for boilers in the cities north of the Huai river, but denied heat to the south cities. The authors show that pollution (measured as *TSP*: total suspended particulates) was larger in the north versus the south of the river.

The authors fit a third-degree polynomial to the data on degrees north of the river (the running variable where the effect is discontinuous at 0 due to the policy), and show Figure 6. Also, they provide details on different functional approximations, varying the polynomial degree from 1 to 5, see Table 7.

Comment on what you think may be the issues with this kind of analysis with respect to the model specification.

Solution. The first thing to notice is that in the graph, if we forget about the fitted third-degree polynomial, it is not very easy to see any large shift in life expectancy. This means that we may have the opposite problem than the one shown in the lecture. Here, fitting a nonlinear model actually shows an effect on life expectancy at the cut-off (zero) but if we had fit a linear model, this effect would not be there. This is confirmed by the results in the table (which authors present as a robustness check). In fact, for a linear or quadratic model, there is no significant effect on life expectancy of being left or right to the cut-off. Only at higher-order polynomials we start seeing an effect.

Exercise 4.9. Show that from the following specification

$$Y_{it} = \beta_0 + \beta_D D_i + \beta_T T_t + \beta_{DT} D_i T_t + U_{it}, \quad \mathbb{E}[U_{it}|D_i, T_t] = 0,$$

we have that

$$\begin{aligned} \beta &= (\mathbb{E}[Y_{it}|D_i = 1, T_t = 1] - \mathbb{E}[Y_{it}|D_i = 1, T_t = 0]) \\ &\quad - (\mathbb{E}[Y_{it}|D_i = 0, T_t = 1] - \mathbb{E}[Y_{it}|D_i = 0, T_t = 0]). \end{aligned}$$

Solution. First, compute conditional expectations exactly,

$$\begin{aligned} \mathbb{E}[Y_{it}|D_i = 1, T_t = 1] &= \beta_0 + \beta_D + \beta_T + \beta, \\ \mathbb{E}[Y_{it}|D_i = 1, T_t = 0] &= \beta_0 + \beta_D, \\ \mathbb{E}[Y_{it}|D_i = 0, T_t = 1] &= \beta_0 + \beta_T, \\ \mathbb{E}[Y_{it}|D_i = 0, T_t = 0] &= \beta_0. \end{aligned}$$

	Linear & Controls	Quadratic & Controls	Cubic & Controls	Quartic & Controls	Quintic & Controls
	(1)	(2)	(3)	(4)	(5)
Panel 1: Impact of "North" on the Listed Variable, Ordinary Least Squares					
TSP (100 $\mu\text{g}/\text{m}^3$)	2.89*** (0.56) [0.988] {492.4}	2.63*** (0.49) [0.068] {489}	1.84*** (0.63) [0.148] {487.2}	1.95*** (0.59) [0.229] {486.3}	1.52** (0.72) [0.671] {487.5}
ln(All Cause Mortality Rate)	0.12 (0.10) [0.276] {39.88}	0.09 (0.10) [0.215] {38.8}	0.26* (0.13) [0.035] {34.11}	0.26** (0.13) [0.908] {35.92}	0.37** (0.16) [0.409] {36.13}
ln(Cardiorespiratory Mortality Rate)	0.13 (0.13) [0.652] {102.3}	0.09 (0.13) [0.243] {101.5}	0.38** (0.16) [0.003] {91.92}	0.39** (0.16) [0.747] {93.34}	0.47** (0.19) [0.696] {94.62}
ln(Non-Cardiorespiratory Mortality Rate)	0.09 (0.10) [0.135] {43.04}	0.05 (0.09) [0.151] {41.27}	0.08 (0.13) [0.933] {43.13}	0.07 (0.12) [0.973] {45.07}	0.19 (0.14) [0.35] {44.97}
Life Expectancy (years)	-1.62 (1.66) [0.101] {757.1}	-1.29 (1.68) [0.6] {758}	-5.52** (2.39) [0.001] {746.8}	-5.67** (2.36) [0.737] {748.2}	-5.43* (2.94) [0.984] {750.2}

Figure 7: Robustness checks of choice of functional form (Chen, Ebenstein, Greenstone, and Li, 2013)

From the third and the forth equation we have

$$\beta_T = \mathbb{E}[Y_{it}|D_i = 0, T_t = 1] - \mathbb{E}[Y_{it}|D_i = 0, T_t = 0].$$

Then, from the first and the second equation we have

$$\begin{aligned}\beta &= \mathbb{E}[Y_{it}|D_i = 1, T_t = 1] - (\beta_0 + \beta_D) - \beta_T \\ &= \mathbb{E}[Y_{it}|D_i = 1, T_t = 1] - \mathbb{E}[Y_{it}|D_i = 1, T_t = 0] - \beta_T \\ &= (\mathbb{E}[Y_{it}|D_i = 1, T_t = 1] - \mathbb{E}[Y_{it}|D_i = 1, T_t = 0]) \\ &\quad - (\mathbb{E}[Y_{it}|D_i = 0, T_t = 1] - \mathbb{E}[Y_{it}|D_i = 0, T_t = 0]).\end{aligned}$$

Exercise 4.10. Show how the coefficients from the interaction model as in Card and Krueger (1993),

$$Y_{ist} = \alpha + \gamma \text{NJ}_s + \lambda_t d_t + \beta_{(1)} (\text{NJ}_s \cdot d_t) + \varepsilon_{ist} \quad (10)$$

correspond to the coefficients in the two-way fixed effects model,

$$Y_{ist} = \gamma_s + \theta_t + \beta_{(2)} D_{st} + \xi_{ist}, \quad (11)$$

and show that $\beta_{(1)} = \beta_{(2)}$ is exactly the same in each specification.

Solution. In (10), we have

$$\text{NJ}_s := \begin{cases} 1, & \text{if } i \text{ is in New Jersey,} \\ 0, & \text{if } i \text{ is in Pennsylvania,} \end{cases} \quad d_t := \begin{cases} 1, & \text{if } i \text{ is in November,} \\ 0, & \text{if } i \text{ is in February.} \end{cases}$$

In (11), we have

$$\gamma_s := \begin{cases} \gamma_{\text{NJ}}, & \text{if } i \text{ is in New Jersey,} \\ \gamma_{\text{PA}}, & \text{if } i \text{ is in Pennsylvania,} \end{cases} \quad \theta_t := \begin{cases} \theta_{\text{Nov}}, & \text{if } i \text{ is in November,} \\ \theta_{\text{Feb}}, & \text{if } i \text{ is in February,} \end{cases},$$

and

$$D_{st} := \begin{cases} 1, & \text{if } i \text{ is in New Jersey in November,} \\ 0, & \text{otherwise.} \end{cases}$$

Computing conditional expectations from (10), we have

$$\begin{aligned}\mathbb{E}[Y_{ist}|s = \text{PA}, t = \text{Feb}] &= \alpha, \\ \mathbb{E}[Y_{ist}|s = \text{NJ}, t = \text{Feb}] &= \alpha + \gamma, \\ \mathbb{E}[Y_{ist}|s = \text{PA}, t = \text{Nov}] &= \alpha + \lambda_t, \\ \mathbb{E}[Y_{ist}|s = \text{NJ}, t = \text{Nov}] &= \alpha + \lambda + \lambda_t + \beta_{(1)},\end{aligned}$$

from which follows that

$$\begin{aligned}\lambda_t &= \mathbb{E}[Y_{ist}|s = \text{PA}, t = \text{Nov}] - \mathbb{E}[Y_{ist}|s = \text{PA}, t = \text{Feb}], \\ \gamma &= \mathbb{E}[Y_{ist}|s = \text{NJ}, t = \text{Feb}] - \mathbb{E}[Y_{ist}|s = \text{PA}, t = \text{Feb}], \\ \beta_{(1)} &= (\mathbb{E}[Y_{ist}|s = \text{NJ}, t = \text{Nov}] - \mathbb{E}[Y_{ist}|s = \text{NJ}, t = \text{Feb}]) \\ &\quad - (\mathbb{E}[Y_{ist}|s = \text{PA}, t = \text{Nov}] - \mathbb{E}[Y_{ist}|s = \text{PA}, t = \text{Feb}]).\end{aligned}$$

From (11), we have that

$$\begin{aligned}\mathbb{E}[Y_{ist}|s = \text{PA}, t = \text{Feb}] &= \gamma_{\text{PA}} + \theta_{\text{Feb}}, \\ \mathbb{E}[Y_{ist}|s = \text{NJ}, t = \text{Feb}] &= \gamma_{\text{NJ}} + \theta_{\text{Feb}}, \\ \mathbb{E}[Y_{ist}|s = \text{PA}, t = \text{Nov}] &= \gamma_{\text{PA}} + \theta_{\text{Nov}}, \\ \mathbb{E}[Y_{ist}|s = \text{NJ}, t = \text{Nov}] &= \gamma_{\text{NJ}} + \theta_{\text{Nov}} + \beta_{(2)},\end{aligned}$$

Combining (10) with (11), we have

$$\begin{aligned}\alpha &= \gamma_{\text{PA}} + \theta_{\text{Feb}}, \\ \gamma &= \gamma_{\text{NJ}} + \theta_{\text{Feb}} - \gamma_{\text{PA}} - \theta_{\text{Feb}} = \gamma_{\text{NJ}} - \gamma_{\text{PA}}, \\ \lambda_t &= \gamma_{\text{PA}} + \theta_{\text{Nov}} - \gamma_{\text{PA}} - \theta_{\text{Feb}} = \theta_{\text{Nov}} - \theta_{\text{Feb}},\end{aligned}$$

and from the expression for $\beta_{(2)}$ from (11),

$$\begin{aligned}\beta_{(2)} &= \mathbb{E}[Y_{ist}|s = \text{NJ}, t = \text{Nov}] - \mathbb{E}[Y_{ist}|s = \text{PA}, t = \text{Nov}] - \underbrace{(\gamma_{\text{NJ}} - \gamma_{\text{PA}})}_{=\gamma} \\ &= \mathbb{E}[Y_{ist}|s = \text{NJ}, t = \text{Nov}] - \mathbb{E}[Y_{ist}|s = \text{PA}, t = \text{Nov}] \\ &\quad - (\mathbb{E}[Y_{ist}|s = \text{NJ}, t = \text{Feb}] - \mathbb{E}[Y_{ist}|s = \text{PA}, t = \text{Feb}]),\end{aligned}$$

and rearranging we see that $\beta_{(1)} = \beta_{(2)}$.